AIA · AGENTIC AI ANALYTICS · METHODOLOGY

# The business-decision study
# Study type 2: a cost model of options

How AIA takes a client's decision from *Zadání* to a report in which every number carries its source and its grade. Version 4, written bottom up from the ChemVitro exemplar and corrected by the external review of 9 November.

THE SPINE · EIGHT STEPS

*Zadání* · *Data* · *Rešerše* · *Předpoklady* · *Model* · *Výsledky* · *Sociomapa* · *Osnova a Zpráva*

The navigator shows nine screens; the outline and the report are one step here. The client's rule is written at the first step and applied, as written, at the sixth. Every number in the report carries its source and one of seven grades. A method the register does not hold stays provisional, and a recommendation that rests on one is *předběžné*.

**Version 4** · 9 November 2026 · working version, for the product owner and the methodology owner
Replaces version 3 (37 pages, 8 November 2026). New: the analytical contract (chapter 2), the cost partition in place of the two weighting families (chapter 12), the convergence rule, oracle-first acceptance, R9 adopted, grades with lineage and validation apart
Source: `prototypes/obchodni-model/methodology/` on branch `chore/obchodni-model-method-v4` · built by `build.js`
Reference report: `prototypes/obchodni-model/exemplar/ChemVitro_2_model-lokality_nigel.pdf`, 44 pages, 5 November 2026
Case pack: `prototypes/case-chemvitro/` (branch `chore/case-chemvitro`) · Design canvas: `prototypes/obchodni-model/`
Agrees with the MVP scope: `prototypes/obchodni-model/MVP-SCOPE.md` as revised for version 2 after the owner's decisions of 9 November 2026

This is design, not product code. Chapter 11 says what exists in AIA today and what does not. Every client, site, person and number in the exemplar is fictional.

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BEFORE THE CHAPTERS

# What this is

On 8 November 2026 the executive chose report 2 of the ChemVitro case study as the report AIA must produce. This document is the methodology behind that report, step by step, and the measure the MVP will be held to.

### The three outcomes of the pivot

| Outcome | Where it lives | Who |
|---|---|---|
| One MVP prototype | The Design canvas *AIA Obchodní model*, `prototypes/obchodni-model/`; thirty-two boards after the sessions of 8 November, Czech, one path from *Zadání* to *Zpráva*; since 8 November the boards carry the ChemVitro case beside the Velis case, each section marked with its case | sessions A–E |
| One methodology PDF | This document, `prototypes/obchodni-model/methodology/`; version 4 after the external review of 9 November and the owner's decisions on it | session F, then the coordinator |
| One MVP scope | `prototypes/obchodni-model/MVP-SCOPE.md` and board MV0; the methodology and the scope must agree | session A |

### What the MVP produces

Study type 2, *Model nákladů lokality*: a decision between options, judged by a rule the client wrote before any result, on a cost model computed by code, with the uncertainty of every input carried through to the verdict. The exemplar is a type-2 report on a fictional client, ChemVitro, choosing among eight sites for a chemical plant or keeping its supplier. Its parts carry the ids R2-P01 to R2-P18 and its tables T1 to T9; this document cites them by those ids.

Types 1 and 3 of the case pack (the situation report without a model; the model beside a worker survey) are not the MVP. The survey lane is frozen (ADR 0046, decision 2).

### How to read it

- Chapter 1 is the spine: the eight steps in one table (the canvas's navigator shows nine; the outline and the report are one step).
- What changed in version 4: an external review of 9 November found the two weighting families did not stop double counting inside a family and let a site lose a weight, read 0,797 against 0,80 on 20 000 draws as a verdict, left the comparables' cost basis and the PERT endpoints undefined, and tested the pipeline only on a case it could pass with wrong arithmetic. The owner decided each point on 9 November; this version writes the decisions in, and marks the two still open (R2-Q02, R2-Q04).
- Chapter 2 is the analytical contract: the defaults every step runs under until the methodologist amends them, and the two release states.
- Chapters 3 to 10 take one step each, all in the same shape: what goes in, what comes out, the checks, how the exemplar did it, what AIA has today, what is new, and what the ChemVitro acceptance expects.
- Chapter 11 says, truthfully, what is built and what is designed. No type-2 code is on `develop`.
- Chapters 12 to 14 cross the steps: the business perspectives and the two Sociomaps; grades, sources and the release gate; governance.
- The appendices hold the glossary, the exemplar in numbers and the index of the canvas boards.

Czech words in italics are terms a person sees on AIA's screens or in the report. English words in `code` are files, modules or plan chunks in the repository. "The review" is the product owner's methodology review of 5 November 2026 (ADR 0046 § The methodology review, R1–R10, K1–K6). "The answer key" is `prototypes/case-chemvitro/ANSWER-KEY.md`. "The scope" is `MVP-SCOPE.md`, session A's document, whose wording this document takes as the source for the perspectives, the chunk ids and the acceptance A1–A6; where the two disagree, the scope wins and the coordinator is told. "The owner" is the product owner; "the methodologist" is the methodology owner, A. Bahbouh.

> **Three things this document does not do.** It does not invent a number: every ChemVitro figure comes from the exemplar's tables or the case pack. It does not decide what waits on the owner or the methodologist; chapter 14 lists those items. It does not describe how a screen works: the boards do that.

CHAPTER 1

# The spine

A type-2 study is eight steps. The client's rule is written at the first and applied, as written, at the sixth. Everything between them exists to make that application honest. The canvas's navigator shows nine screens: the outline and the report are one step, because the report's structure is built once and printed once (the scope, § 1).

| Step | Goes in | Comes out | The check | In the exemplar |
|---|---|---|---|---|
| **1** ***Zadání*** | the decision, its options and constraints, who decides and when, the client's files | the rule with its conditions, frozen and dated; interpretations recorded | frozen before any comparative result; every condition has a threshold and a measure | eight sites and the buy baseline; the CFO's rule of 2. 11. 2026 with four conditions; one interpretation waits for the CFO (R2-P02) |
| **2** ***Data*** | every client file and document | a contract per file; a findings table with a disposition per finding | units, currencies, price years, duplicates, missing, outliers, quotes that do not cover the same thing | 42 findings; the key one is the Hanshin price (R2-P04 § 3.4, P1) |
| **3** ***Rešerše*** | what the files cannot give: public prices, rates, taxes, limits, law | typed public inputs, each with a quote, a source, a date and a state | verified at the source, or marked *prozatímní* | 16 public inputs unverified of T9's 67 rows (the text says 54); *Stav podkladů* says so on page 1 (R2-P01) |
| **4** ***Předpoklady*** | the contracts, the public inputs, the client's answers | one table of inputs: low, likely, high, distribution, source, grade, guarantor, state | estimated from client data where possible; the scope of every estimate checked; correlations declared with a stability test fixed in advance | 67 inputs (T9); overrun and delay from 36 comparable plants, drawn together (R2-P05) |
| **5** ***Model*** | the input table, the options, the timeline | the base case and the seeded futures; the same bytes on a rerun | one model for every option; the convergence rule of chapter 2; one change at a time in the extra computations | cost of supply 2027–2044; 20 000 futures, seed 20270201 (R2-P04 § 3.3, Appendix B) |
| **6** ***Výsledky*** | the futures and the base case | the rule applied as written; what could change the verdict; value of information; finance, operations, schedule | the verdict shown as a boundary with its resolution state; no probability outside the model; information valued for the rule first, through declared answer branches | no site passes; the flip at 14/15/16 % compensation chance; EVPPI second (R2-P06–P11) |
| **7** ***Sociomapa*** | the options' results across the futures, or their weighted scores across scenarios | a map with a mode contract, its relation matrix and what it may claim | links only where the contract allows; a stability test declared in advance; relative apart from absolute | the futures map failed its stability test and the report says so (R2-P12, T7, T8) |
| **8a** ***Osnova zprávy*** | every claim the report will make | an outline ordered by the argument; each point with its source or method and why it matters | no point without a source; a point that is a judgement is marked as one | the part ids R2-P01…P18 and the grade legend (R2-P04 § 3.5) |
| **8b** ***Zpráva*** | the closed outline | the report: summary, chapters, limits, literature, appendices, tables | every number with a grade in small type and a numbered source at the chapter's end; a validity review before approval | 44 pages, seven grades, *Meze studie* admits what is unverified (R2-P15) |

## 1.1 The shape of each step chapter

Chapters 3 to 10 use seven parts, always in this order and under these names, so that two steps can be compared line by line.

| Part | What it answers |
|---|---|
| What goes in | The inputs the step takes, and from which earlier step or from the client. |
| What comes out | The artefact the step produces, as the next step and the report read it. |
| Checks | What must hold before the step is closed. A check is a refusal or a record, never advice. |
| How the exemplar did it | The ChemVitro report's own practice, cited by part id and table. |
| In AIA today | The code on `develop` the step reuses, by module. Nothing listed here is type-2 code; it is what the pivot stands on. |
| New | What has to be built, with the board and the plan chunk that build it, as the scope names them (`plan#CHUNK`; a chunk marked **new** exists in no plan file yet and the scope defines it in one line). |
| ChemVitro acceptance | What the answer key planted for this step, which of the scope's acceptance lines A1–A6 it serves, and which of the review's criteria K1–K6 it answers. |

## 1.2 Acceptance and waves

Acceptance is in two layers since version 4 (the owner, 9 November, R2-Q11). First an independent numerical oracle: toy fixtures with known answers (O-CF for the cash flows, O-INFO for the value of information) and a frozen ChemVitro configuration (O-CHEM), written by an analyst session that never writes the model code and signed by the methodologist (chapter 14). Second the answer key, as the scope (§ 6) wrote it, in six lines: A1 the three traps priced; A2 the EPC exclusions, the overrun from the comparables, spot purchases while late, tax reliefs out of the pre-tax rule, the kilogram and USD rows read right; A3 the rule applied as written at 8, 6 and 10 %, with the boundary printed where a declared judgement moves the verdict; A4 the renewal and Hanshin prices named as the assumptions most worth checking; A5 the report's structure R2-P01…P18 with every number graded and sourced and the tables from the same computation; A6 the same bytes on a rerun and a changed input refused without re-running downstream. The five waves of ten weeks (scope § 7, resourced on 9 November: three build sessions, one analyst session, the methodologist one day a week, 97–139 session-days in all) are W1 contracts, oracles, the study shell, the rule freeze and the first data contract; W2 reconciliation, bindings and the cash-flow model; W3 finance, joint uncertainty and the rule with its convergence; W4 information value, the partition and both maps; W5 the report, invalidation, acceptance and the independent review. Each ends in a slice that runs on the ChemVitro files with nothing paid. Week 10 may produce a reviewed *předběžné* report; the calendar does not turn an unresolved comparison into an approved recommendation.

## 1.3 The rules the whole spine keeps

- **The client's rule decides.** It is written before any result and judged as written (K3). Anything else the study computes is evidence for "what could change the conclusion", not for the recommendation (review R3, R10).
- **A number is a reference to its origin.** It carries a source, a method, an uncertainty and a validation status (review I3). A number the origin does not support is not printed.
- **Unknown is never good.** A missing input keeps a range; a gate that is unknown blocks the recommendation until it is evidenced (review R2). A probability is a share of the model's futures, never a claim about the world.
- **Models propose, code computes, a person approves.** AIA's standing rule (ADR 0046 decision 1). A language model drafts text and suggests ranges; every number in the report comes from a program that gives the same bytes on a rerun.
- **A method the register does not hold stays provisional.** Its results print *odhad (metoda čeká na schválení metodikem)*, and a recommendation that rests on it stays *předběžné doporučení*: document approval and recommendation eligibility are two states (R9, adopted 9 November; chapter 2).
- **A comparison is resolved or it is an interval.** The futures double from 20 000 at declared checkpoints until each gate and decisive comparison clears its bound; one that does not prints as unresolved (chapter 2, chapter 8).
- **Changes propagate.** A revised input marks every downstream result and approval stale until someone reviews it (K4). Every extra computation changes one assumption at a time and says so (R2-P15).

CHAPTER 2

# The analytical contract

Seven decisions have to be made before any model code is written, because each of them changes what a number in the report means. Version 4 fixes a default for every one of them. The owner adopted the seven as the baseline on 9 November 2026; the methodologist reviews them on his first day and changes what he disagrees with. Until he does, the defaults bind every step.

## 2.1 The seven defaults

| Decision | Who decides | Default | What breaks if it stays open |
|---|---|---|---|
| **Comparable-cost basis** | methodologist sets the basis; owner approves the business assumption; code enforces it | For every comparable: the original EPC estimate, the actual expenditure on the same scope, the exclusions, currency and price year. Overrun = actual like-for-like EPC ÷ original estimate − 1. Site exclusions (spur, bridge, pre-treatment) are added once, never inside the overrun as well. The fictional case's footnote is clarified before its percentages are read as this ratio (open, R2-Q02; chapter 6). | The investment and the 175 mil. EUR gate have no defensible basis; the spur can be counted twice. |
| **Complete parameter manifest** | code owns the schema; methodologist approves uncertainty semantics; owner or client approves business assumptions | Every value the model reads is a row: fixed values, ranges, structured tables, taxes and depreciation, tax capacity, timing dependencies, distributions, cross-option dependence, scales and method settings. Each row has unit, period, source locator, the grade attributes, an owner and its dependencies. A bound without a justification blocks every numerical claim that reads it. Counts in the report (inputs, public inputs, judgements) come from the manifest, never from the text. | Hidden defaults escape provenance and invalidation; the text says 54 where the table has 67. |
| **Rule freeze and amendments** | owner or client confirms the meaning; code records and enforces | Objective, gates, rates, budget currency and year, the investment reading and tie treatment are frozen before any ChemVitro model result. For the fixture the investment condition reads as total base-case installed investment in 2026 EUR (the owner, 9 November, R2-Q01; Aram confirms as the case's author). Exposure to earlier results is recorded. An amendment is a new revision with reason, approver, a rerun and the old and new outcomes disclosed. | There is no authorised rule to apply; a result can be rationalised by a rule written after it. |
| **Distribution endpoints** | methodologist; owner or client supplies the business judgements | A hard bound defines the support; an elicited quantile is fitted as a quantile, never read as a bound. Reconciled observations keep their joint structure (the comparables' overrun and delay drawn together). Disagreement stays as separate scenarios or an approved mixture. Missing evidence keeps a justified range, never an automatic midpoint or a made-up distribution. | Tail risk acquires a precision nobody stated; truncation hides the cases the gates exist for. |
| **Numerical convergence** | methodologist approves the error control; code chooses batching and resource limits | Start at 20 000 independent joint futures; double at declared checkpoints, capped at 1,28 million. Each gate, median comparison and decisive information comparison gets a repeated-look-valid exact binomial bound (order-statistic bounds for quantiles), with a total numerical error budget of 5 % across comparisons and looks. A bound that overlaps the threshold means *unresolved*, and the report prints the interval. ±0,5 percentage points alone certifies nothing. Medians are compared as medians, not as the median of paired differences. | A seed or a resource cap becomes the decision rule; 0,797 against 0,80 on 20 000 draws is read as a verdict (chapter 8). |
| **Enquiry updating** | methodologist approves the update; owner or client approves what counts as evidence and what action follows | Every enquiry declares its answer branches, their reliability, cost and duration ranges and the dependencies it shares with other inputs. The joint model is conditioned on the answer, the timetable shifts by the duration, the cost of asking enters, and the rule is reapplied. A letter from an authority does not remove all uncertainty by itself. | The value of information cannot support an action or a deadline; the enquiry table is a wish list. |
| **Release eligibility** | owner sets the policy; methodologist registers methods; code enforces eligibility | Two states, kept apart: **document approval** (this version was reviewed and may leave AIA) and **recommendation eligibility** (the recommendation may be stated as approved). Labelled unregistered estimates are allowed internally and in an independently reviewed *předběžné* client draft. An approved recommendation needs registered methods, the independent review and every decisive comparison resolved (R9, adopted by the owner 9 November, R2-Q05; chapter 13). | "Approved" means two different things to the client and to the methodologist. |

Sources: the exemplar's rule and its pending interpretation (R2-P02 § 1.3); the cash-flow convention (Appendix B of the exemplar); the enquiry limits (R2-P09 § 8.3); the comparables' footnote (`srovnatelne_zavody.xlsx`, A40). The seven replace the implicit EPC uplift, the fixed N of 20 000 as an acceptance criterion and the ambiguous release warnings of version 3.

## 2.2 The two release states

| State | Granted when | What the report may say |
|---|---|---|
| **Internal draft** | composed by code from a stored run | everything, labelled; never leaves AIA |

| State | Granted when | What the report may say |
|---|---|---|
| **Approved document, *předběžné*** | the independent review (chapter 14) is recorded with its open items; every number carries its grade; unregistered methods are labelled | a provisional recommendation with its qualifications named; no "doporučujeme" without the qualifier |
| **Approved document, recommendation eligible** | every method the recommendation rests on is in the register; the review's items are closed; every decisive comparison is resolved under the convergence rule | the recommendation as approved investment advice |

Document approval is what `application/report_approval.py` records today, once per stored version. Recommendation eligibility is new: a field of the approval computed by code from the register, the review record and the comparisons' resolution states, never typed.

## 2.3 Who does what in the ten weeks

Three build sessions, one analyst session that never writes model code and freezes the oracle fixtures (chapter 14), and the methodologist one day a week, starting with the fixtures and these seven defaults (the owner, 9 November, R2-Q07 and R2-Q12). The owner and the methodologist decide; a decision that has not arrived blocks the dependent result, not the engineering around it.

CHAPTER 3 · STEP 1

# *Zadání*: the decision and its rule

The study begins by writing down what will be decided, between which options, under which constraints, by whom and when, and by which rule. The rule is frozen before anyone sees a comparison of the options.

## 3.1 What goes in

- The client's brief, as typed or pasted: one field of free text and the files that come with it (the case pack's brief is in `CASE.md` § 3).
- The decision-maker and the date the decision is taken, and the date the study must be ready.
- The options, each with the client's source for it, and a baseline every option is compared with. For ChemVitro: eight sites from the agencies' offers and "keep buying" from the supplier's letter.
- The constraints every option must meet (investment ceiling, start date, one plant at one site).
- The client's answers to the study's questions. The case pack fixes them in `CASE.md` § 8; where an answer is "we don't know", the input becomes a judgement with a range and says so.

## 3.2 What comes out

- **The decision record:** what is decided, why now, the goal, who decides, by when. The exemplar's § 1.1–1.2 (R2-P02).
- **The options and the baseline**, named as the client names them, with the source of each.
- **The rule**, frozen and dated, in the client's words: the measure it reads, the choice it makes, and each condition with its threshold. ChemVitro's: the highest median saving against buying, NPV of the cost of supply 2027–2044 at 8 %, if P(beats buying) ≥ 0,75, P(loss > 20 mil. EUR) ≤ 0,10, P(production by 1. 7. 2030) ≥ 0,80 and investment ≤ 175 mil. EUR; else recommend nothing and name the assumption most worth checking (R2-P02 § 1.3).
- **Interpretation records.** Where a condition does not say how it is measured, the study writes its reading down, the alternative reading, what each would change, and who confirms it. The exemplar's condition (4) names no probability; it is judged on the base case, the stricter reading "in every future" would fail all eight sites without changing the verdict, and the CFO is asked to confirm (R2-P02 § 1.3).
- **The views the client asked for beside the rule:** after-tax NPV, the verdict at 6 % and 10 %. They are shown; the rule does not read them.
- **The analyses the report will carry** (board Z4) and the perspectives that apply to the case (chapter 12).

## 3.3 Checks

- The rule exists before the first comparative result, and the freeze is dated and attributed. Reading the client's files at intake is feasibility work and is recorded as such (review I2).
- Every condition has a measure the model can compute and a threshold. A condition without one gets an interpretation record, never a silent default.
- An amendment of the rule after the freeze is approved by the decision-maker, dated, and printed in the report (board Z3, *zámek*).
- Every option is described comparably: the same plant, the same horizon, the same discount rate (review R1, question 1).
- The brief is Class B material by rule (`domain/material_rules.py`); a language model may read it, a client's dataset is Class A and is not sent to a model.

## 3.4 How the exemplar did it

The report states the decision, the eight sites, the fixed plant (40 000 t a year, 95 FTE, one process design) and the dates: the board meets on 29. 1. 2027, the site is decided on 1. 2. 2027, production by 1. 7. 2030, investment at most 175 mil. EUR (R2-P02 § 1.2). It prints the rule in full and says why its date matters: "that the rule was written before the computation prevents the result from being justified afterwards by a rule that would suit it" (§ 1.3). It records the one interpretation, prices the alternative reading and asks the CFO. It also notes that the rule reads the pre-tax NPV and that after-tax figures and other rates are shown at the client's request.

## 3.5 In AIA today

- The study's working copy with revision-checked saves and the operation log (`application/working_copy.py`, `domain/edit_ops.py`, `infrastructure/study_workspace_repository.py`), and immutable Design Revisions (`domain/design.py`, `infrastructure/study_design_repository.py`). The frozen rule is a revision of this kind.

- The Brief stage of the research workbench (`apps/web/src/components/rehome/research/`, `BriefStep`) and its materials (`domain/brief_materials.py`), reused as the screen the brief is typed into.
- The Lead, which triages a brief and proposes edits a person accepts (`application/research_lead.py`, ADR 0026, ADR 0041).
- The approval ledger with its closed set of subjects (`domain/approval_ledger.py`): the rule's freeze and every amendment become ledger rows.
- Material classes by rule (`domain/material_rules.py`, ADR 0008).

## 3.6 New

- A study kind for a business decision, `StudyKind.BUSINESS_MODEL` (ADR 0044; `cost-model-study#CP3`), and the business *Zadání* as data on boards Z1–Z7: the eight parts (`decision-brief#DB0`), triage as a Lead turn (`#DB2`), the decision core and the rule (`#DB3`), readiness R1–R7 (`#DB4`), the hand-off naming the files each step takes (`#DB5`).
- The rule as a typed object of n conditions, each with its own threshold and, where stated, a probability, judged per option at the stated rate and two alternates, frozen as a Design Revision (`decision-brief#DB3b`, **new**, extending `cost-model-study#CR7`, which judges one quantity against one threshold). Board Z3 § 03b (session B) shows it for ChemVitro: each condition with its threshold and what it reads, the four conditions named as gates that no saving outweighs, and the lock line (frozen 2. 11. 2026, amendments 0, interpretations 1, awaiting confirmation).
- The interpretation record a named person confirms: what the rule does not say, the chosen reading, the other reading and its effect, who wrote it and who confirms, where the report prints it (`decision-brief#DB3b`; board Z3 § 03b, *Záznam výkladu*).
- The model's parts as data on board Z5 § 04b (session B): the target quantity, the baseline, the futures, the shared timeline, what is in and outside the model, the investment ceiling as a gate, the result views the model will feed, and the perspectives that have components in the case.
- The perspectives chosen for the case (chapter 12), each with its leaves, in *Zadání* rather than in the model.

## 3.7 ChemVitro acceptance

- The rule is the CFO's of `CASE.md` § 6, with all four conditions and the "recommend nothing" clause, dated before any result (MV0 M-4; K3).
- The investment condition has an interpretation record, and the record names the CFO as the person who confirms it.
- The baseline is "keep buying" at the supplier's terms, not a site.
- The study asks the client only what `CASE.md` § 8 answers; anything else is a judgement with a range, labelled *úsudek*.

CHAPTER 4 · STEP 2

# *Data*: the client's files under contract

Every client file is read, checked and bound by a contract before a number from it reaches the model. The checks produce findings, and every finding gets a disposition the report can print.

## 4.1 What goes in

- The client's data files: for ChemVitro ten files in `data/`, from monthly purchases to air-quality measurements (`CASE.md` § 9). The worker survey belongs to type 3 and is not read.
- The client's documents: seven PDFs, from the board's brief to the consultant's memo on air permitting. Each number taken from one is cited with its page.
- The decision record from step 1, which says what each file is for.

## 4.2 What comes out

- **A data contract per file:** which columns are used and how; units, currency and price year per column; which columns are personal and never leave the analysis copy; the period covered (board MV1; `client-data-in-analysis` chunk 2).
- **A quality report per file**, accepted by a person before use: missing, duplicate and out-of-range counts per used column over every row (`client-data-in-analysis` 2a).
- **The findings table with dispositions.** Every finding of the checks, with one of four dispositions: used; corrected, with the reason; left out; taken over as an assumption with a range. The findings that change the result or its reading go into the report (the exemplar's P1, R2-P18 Appendix B).
- **A document register:** each document by the name the report cites it under and its file name (the exemplar's Appendix A).

## 4.3 Checks

The checks are the exemplar's list (R2-P04 § 3.4), made explicit:

| Check | What it looks for | ChemVitro instance |
|---|---|---|
| Units | a row whose unit differs from its column's | July 2025 volume 2 844 000: kilograms, read as 2 844 t (P1) |
| Currencies | mixed currencies in one column; a currency column that contradicts the header | comparable plants in EUR, CZK and PLN; one purchase row in USD (answer key § 3) |
| Price years | amounts in different years' prices without a year | one comparable plant without a price year (P1) |
| Duplicates | the same entity twice | one comparable plant entered twice; counted once, 36 plants (P1) |
| Missing values | cells the contract expects and does not find | no start date of the energy price fixation; waste prices only to 2028 (R2-P01 *Stav podkladů*) |
| Outliers | a value outside the range the file itself declares | one plant with capacity 91 kt against 18–60 kt; used only for overrun and delay (R2-P15) |
| Scope of a quote | quotes that do not cover the same thing | a cheaper PL2 waste quote covers only non-hazardous waste; PL1's rail rate assumes a spur that does not exist (answer key § 3) |
| File against brief | a number the brief states and the file contradicts | "about 1 470 EUR/t" against a 2026 average of 1 694 and 1 868 in September 2026 (P1) |
| Personal data | a column holding a person's identifier | none in type 2's files; the survey's respondents are pseudonymous and belong to type 3 |

A contract that names a column the file does not hold, or a quality report the person has not accepted, blocks the file from the model (MV0 M-1).

## 4.4 How the exemplar did it

The model uses every client file except the worker survey and all seven documents (R2-P04 § 3.4). The checks produced 42 findings, each with a recorded decision; the ones that change the result are in table P1 (Appendix B): the Hanshin price modelled as a range 1 470–1 694–1 868 EUR/t; the kilogram row; the single spot purchase of February 2026 turned into a spot premium of 20–60 %, most likely 39 %; the duplicate, the missing price year and the 91 kt outlier among the comparable plants; consumption growth of 4 % in the data against the client's 1–3 %, the client's range used; raw-material freight not added because the raw-

material price is delivered to site; Odrzyca's rail quote and Eschfeld's bridge. Twenty-two inputs the files do not hold became judgements with a range. A separate document, *Kontrola dat*, holds the contracts and all 42 findings (Appendix C); the report prints the ones that matter.

### 4.5 In AIA today

- A client's sources kept once per client and SHA-256, read by a format reader into segments with locators (`domain/client_sources.py`, `infrastructure/source_reading.py`, `infrastructure/client_source_repository.py`): PDF by page and table row, XLSX and CSV by sheet and row, DOCX by paragraph.
- The dataset preview: per column the header, kind, fill, distinct values and samples, with personal columns detected and masked (`domain/client_datasets.py`; patterns pinned in `fixtures/client_datasets/personal.json`).
- The intake sheet, *Přidat podklady*: each file kept and read at once, one knowledge item per file (`apps/web/src/components/aia/clients/IntakeSheet`).
- Document text with ZIP bounds and a bounded table reader (`infrastructure/document_text.py`).
- The evidence policy for client data: client-reported evidence under an accepted contract (ADR 0027, ADR 0037; `client-data-in-analysis` 8a built).

### 4.6 New

- The data contract as a type and the route a person accepts it through (`client-data-in-analysis#2`), the quality report (`#2a`), the analysis copy of named columns and its accessor (`#3`, `#3a`, `#3b`), admission as *z dat klienta* (`#5`), the contract accepted on the Data step (`#10`). Boards MV1 § 01b–03b and M1 § 01b–03b (session B) show it for the purchases file: the contract per column with the check note, the quality report with a disposition per row, what the model gets from it (summaries, never rows), and the 42 findings with the seven that change the result.
- A number from a document proposed with its page and checked literally on that page (`client-data-in-analysis#11`); an estimate from client rows as a Metodika method (`#12`).
- Declared conversions (unit, currency at a dated rate, price year), each a finding with its disposition, and the P1 table of findings with the four dispositions *použito / opraveno / vynecháno / převzato jako předpoklad* (`client-data-in-analysis#13`, **new**; board M1 § 03b). `cost-model-study#CR1` covers reconciliation, period coverage, currency and robust outliers, accepted on record.
- The document register with cited names (the exemplar's Appendix A; part of `decision-brief#DB5`'s hand-off).

### 4.7 ChemVitro acceptance

The answer key planted these in the files (§ 3). A study that misses one has a gap in its checks, whatever its verdict.

- `srovnatelne_zavody.xlsx`: P-17 twice, the second with a rounded investment; P-23's capacity 91 in t/day, not kt a year; P-31 without a price year; investments in EUR, CZK and PLN in each plant's own price year.
- `nakup_vtx40_2023-2026.xlsx`: the 2025-07 volume in kg; the 2024-03 price in USD; an extra 2026-02 row from another supplier, the spot price for a late start. The key names 2 180 EUR/t and the exemplar 2 320 against 1 672 (P1); both are the same row read two ways, price alone or price plus freight (2 180 + 140 against 1 576 + 96), as board MV1 § 02b shows. The acceptance reads the file and says which it prints.
- `trh_prace_regiony.xlsx`: wages gross, in local currency, without employer contributions.
- `odpady_nabidky.csv`: PL2's second quote covers only non-hazardous waste.
- `doprava_nabidky.csv`: PL1's rail rate assumes a spur to the site.
- `hodnoceni_lokalit.csv`: one rater scored with the scale reversed and says so in a note; the consultant adds about 2,5 points to the two sites its agency brokered; two cells are missing. Type 2 does not use this file in its model; if it draws an expert map, the map must show these.

Criteria: K6, claim integrity, where a table and the text disagree about a reading; K4, since a corrected finding propagates to every result that read it.

CHAPTER 5 · STEP 3

# *Rešerše*: public inputs with a state

What the client's files cannot give, public sources give: prices, rates, taxes, limits and law. Each is a typed input with a quote, a source and a date, and it stays provisional until someone has verified it at the source.

## 5.1 What goes in

- The list of public inputs the model needs, from the decision record and the contracts. For ChemVitro, `CASE.md` § 10: the EU ETS allowance price and its outlook and whether a 24 MW boiler falls under the scheme; air-quality limit values and the permitting rules of three countries; hazardous-waste fees; wages, employer contributions and unemployment; corporate income tax, depreciation and the nature of each incentive; industrial energy price outlooks; exchange rates.
- The client's data posture (*Pravidla*, `domain/client_rules.py`) and the study's budget, which bound what a priced route may spend.
- Findings that frame the problem: the literature the report cites (R2-P03), found and quoted the same way.

## 5.2 What comes out

- **Typed public inputs** (board MV2): a value with its unit, the quote it was read from, the source and its date, the kind of source (*Úřední / Oborový / Ostatní*), the grade (*veřejný zdroj* or *právní předpis*) and the state: *ověřeno* once checked at the source, else *prozatímní*.
- **Findings by analysis** (board MV2b): PESTEL, Porter or benchmark findings as proposals a person accepts, each a quote in a captured source.
- **The literature review** as the report prints it: for each work, what it says and what follows for this report (R2-P03, five works).
- **The state line for the report's first page:** how many inputs are public, how many unverified, and whether the unverified ones could change the verdict (R2-P01 *Stav podkladů*).

## 5.3 Checks

- A finding is a quote in a captured source, or it is nothing (ADR 0017). A value without a quote and a date is not an input.
- A query's data class is inherited from what it was written from and never lowered by its words (ADR 0017, `domain/deep_research/classification.py`). A client's dataset never leaves as a query.
- A provision is cited; no legal advice is given (`CASE.md` § 10, R2-P04 § 3.3).
- An unverified input is marked *prozatímní* in the input table, in the chapter that uses it and on page 1. Verification at the source is a dated act by a named person and changes the state, nothing else.
- A public input the exemplar's method tests for sensitivity says whether its range changes the verdict (chapter 8).

## 5.4 How the exemplar did it

Sixteen inputs are public (the text says "16 of 54"; T9 has 67 rows, a claim-integrity finding recorded in chapter 6): the allowance price (89,10 EUR/t on 28. 10. 2026, Investing.com, unverified), its growth from December 2030 futures, free allocation, energy price volatility and long-term movement, three price correlations, the CZK and PLN rates, employer contributions in three countries, inflation (T9). The report says, on page 1, that the authors could not verify them at the source because access to public sources was closed while the report was written, that the amounts may change after verification, and that the correlations and the energy trend do not change the verdict while stricter 2030 air-quality limits could (R2-P01). The tax rates, depreciation, the form of incentives and Directive (EU) 2024/2881 are likewise provisional; the incentives are read correctly as tax reliefs in Czechia and Poland and a cash grant in Germany (R2-P10). The literature chapter cites AACE 18R-97, Flyvbjerg, Merrow, Vose, Strong and Oakley, Williamson and Lempert, Popper and Bankes, each with the consequence for the report (R2-P03).

## 5.5 In AIA today

- Deep Research, frozen since #166: plan, investigate, merge, verify, synthesize, publish; grounding of every quote in a content-addressed snapshot of the same track (`domain/deep_research/`, `grounding.py`, `web.py`, `sources.py`); the sealed bundle.
- The one way a query or URL leaves: `application/web_retrieval.py` `RetrievalGate` (classify, egress, metering, reserve, journal, call, journal). Live routes: Czech and English Wikipedia, priced Brave, Common Crawl (`aia_executors.deep_research_runtime`).
- Design Research and Interpretation Research as the two purposes; a finding never writes a design or a deterministic artefact (ADR 0021).

- A captured source a person checked becomes a SOURCE knowledge item, *Ověřeno, přidat zdroj* (`application/verified_sources.py`).
- The Lead's web lookup, accepted and not built (ADR 0040).

### 5.6 New

- Typed inputs (value, unit, period, range, source, as-of date) extracted by code from accepted quotes and from structured routes, with the state *prozatímní / ověřeno u pramene* (`deep-research-v2#DRV1` the typed input, `#DRV2` the extractor, `#DRV3` the extraction step); credibility tiers *Úřední / Oborový / Ostatní* (`#DRV5`); as a provenance of a model input, `cost-model-study#CP1`'s `public_source`. Board MV2 § 01b (session B) lists the sixteen provisional rows of T9 with value, unit, source, date, grade and state, and § 02b makes verification a step with three outcomes: Deep Research finds and cites (*prozatímní*), the researcher opens the source and records the date and their name (*ověřeno u pramene*), what no source holds stays a judgement with a guarantor.
- Structured public routes: SDMX for Eurostat and the ECB (`public-data-routes#PDR1`), ČNB rates (`#PDR2`).
- A legal point as a verbatim provision that becomes a parameter (tax rate, depreciation, contributions): e-Sbírka and EUR-Lex routes, the legal point, legal parameters (`legal-context#LGL1`, `#LGL2`, `#LGL3`).
- The first-page state line composed from the input table's states (part of the report-2 recipe, `research-findings-stage#FND13`, chapter 10).
- Findings by analysis (PESTEL, Porter, benchmark) as proposals, each a quote (board MV2b, unchanged in this round).

### 5.7 ChemVitro acceptance

- Every public value in the model carries a quote, a source, a date and a state; none is invented (`CASE.md` § 11).
- The 24 MW boiler is above the 20 MW threshold: carbon is a cost at every site (answer key § 3, memo p. 1).
- Incentives are read by their nature: the Czech investment incentive and the Polish CIT exemption are tax reliefs and do not change the pre-tax figure the rule reads; only the German GRW is a cash grant. Subtracting PL2's "CIT exemption up to 50 %" from the investment is a misreading (answer key § 3).
- The revised directive (EU) 2024/2881 with its 2030 limits is noted as a caveat, as the exemplar does; it is not planted and earns credit, not a requirement.
- Criteria: K2, where a public range is a decisive input; K5, where verification at the source has a date that the plan of action depends on.

CHAPTER 6 · STEP 4

# *Předpoklady*: one table of inputs

Every input of the model is one row of one table: its low, likely and high value, its distribution, its source, its grade, the person who stands behind it and its state. The table is the contract between the data and the model, and the report prints it whole.

## 6.1 What goes in

- The data contracts and the findings table from step 2; the typed public inputs from step 3; the client's answers from step 1.
- The comparable cases the client holds, from which ranges can be estimated rather than judged: for ChemVitro, 36 comparable plants with their overrun, delay and what each built beyond the plant.
- The scope statement of every estimate the client supplies: what the EPC contractor's figure includes and excludes (`epc_odhad.pdf` p. 3).

## 6.2 What comes out

**The input table**, the exemplar's T9 (R2-P18). Its columns are the method:

| Column | Rule |
|---|---|
| *Vstup, jednotka* | one input, one unit; a fixed parameter is a row like any other |
| *Dolní · nejspíš · horní* | three values; equal for a fixed value. What the ends mean is written down for every decisive input (review R6), and the meaning decides how they become a distribution (chapter 2): a hard bound (contractual, physical) defines the support; an elicited quantile (a 90 % interval, a credible worst case) is fitted as a quantile and never read as a bound; a missing fact keeps a justified range, never an automatic midpoint |
| *Rozdělení* | PERT by default for a range a person stands behind, its endpoints read as the row says; the estimate's own distribution where it comes from data; an empirical draw from comparable cases that keeps their joint structure; yes/no with a probability for an event; disagreement kept as separate scenarios or an approved mixture; a fixed value (*pevná hodnota*) |
| *Zdroj* | the file and row, the document and page, the public source and date, or "not in the files" |
| *Stupeň* | one of the grades of chapter 13: *z dat klienta*, *odhad*, *úsudek*, *veřejný zdroj*, *právní předpis* |
| *Stojí za ním* | the guarantor: a named role in the client (the CFO, head of purchasing, head of EHS) or the author. Every assumption has an owner (the review's "to keep") |
| *Stav* | *ověřeno* (from client data), *prozatímní* (public, unverified) or *úsudek* (the client has no answer) |

Beside the table: **the correlation record** (which inputs move together, with what coefficient, why, and the stability test their judgement will be put to), **the scope check** of every supplied estimate against each option's documents with the exclusions priced, and **the stress cases** outside the ranges, kept apart from the futures (board M2 § 04).

## 6.3 Checks

- Where the client's data allows an estimate, the range comes from the data, not from a judgement; the row says which. The exemplar: overrun and delay from the comparable plants, not from the estimator's ±25 % (R2-P05).
- Two things that happened together in the data are drawn together in the model. Overrun and delay have a Spearman correlation of 0,89 across the comparable plants, so each future draws one real plant and takes both (R2-P05).
- Every supplied estimate is checked for scope against each option's documents, and each exclusion is priced or stated as missing (R2-P05, fourth assumption; answer key § 3: a study that adds nothing to the EPC figure has missed all three traps).
- **The comparables' cost basis is reconciled before their overrun is used** (chapter 2). The file says totals are actual costs "including off-site infrastructure" and overruns are "against the EPC estimate at decision"; applying that overrun to ChemVitro's EPC figure and then adding the spur, the bridge and pre-treatment can count them twice. The overrun is defined as actual like-for-like EPC scope over the original estimate, the exclusions are added once, and the fictional footnote is clarified first (open, R2-Q02; the recommendation is a versioned clarification of `srovnatelne_zavody.xlsx` A40). The cohort is checked for comparability (capacity 18–60 kt, the duplicate plant P-17 removed, the 91 kt outlier treated by a declared rule).
- **Every value the model reads is a row of the manifest** (chapter 2): taxes and depreciation, the incentive's form, the directive's limits and the method settings included. The counts the report prints come from it.
- A decisive input has an uncertainty type: natural variation, parameter, expert disagreement, model structure, or a severe event outside the range; the last goes to a stress test, not into the distribution (review R6; board M2 § 02).
- Experts who disagree from the same documents are a mixture that does not narrow the range; the report says they differ in reading, not in facts (board M2).

- A correlation that is a judgement is declared as one, with the alternative settings it will be tested under, before anything is computed (R2-P05 fifth assumption, Appendix B).
- No input the files do not hold is given a point value. It gets a range and the grade *úsudek*; the exemplar has 22 such inputs (R2-P18 Appendix B).

### 6.4 How the exemplar did it

Chapter 4 (R2-P05) names the two assumptions the verdict rests on that no document supports: the 20 % chance that an authority imposes compensation measures where the station is under the limits, and the terms of purchase after 2035. Then three more: overrun and delay from 36 comparable plants (median overrun 17 %, 90th percentile 30,2 %, maximum 61 %; delay median 3 months, maximum 15), drawn jointly; the exclusions of the EPC estimate priced per site from the site's documents (18 ha of land everywhere; the bridge at Eschfeld 3,2–5,0 mil. EUR; the spur at Odrzyca 3,4 km at 1,9–2,8 mil. EUR per km; pre-treatment at Lipová and Lindenau 6,7–14,1 mil. EUR; compensation 3–15 mil. EUR where imposed; the park fee at Wendelstedt 1,1 mil. EUR a year); and the energy price correlations as judgement (0,8 electricity between countries, 0,6 electricity–gas, 0,4 gas–allowance), with the stability test declared in Appendix B. The base case already carries the median overrun. The report's text says the model has 54 inputs and that 16 of 54 are public; the table T9 has 67 rows (29 *ověřeno*, 16 *prozatímní*, 22 *úsudek*). Session B counted from T9 and recorded the difference on board M2 § 01b as a finding for the report; it is the kind of contradiction between text and table that K6 exists to catch, and this document counts from T9.

### 6.5 In AIA today

- Nothing holds an input table yet. The nearest ancestors are the simulation core's frozen WorldModel with reject-not-clip validation (`domain/simulation/`) and the research template's readiness rules (`domain/research_design.py`), which show the pattern: a typed object, validated, pinned to a run.
- Guarantors and audit: the approval ledger (`domain/approval_ledger.py`) and the audit trail every admin route writes.
- Methodology as data the methodologist owns: versions, activation and bounds per editable field (`domain/methodology/`, ADR 0023). The distribution defaults and the stability thresholds belong there.

### 6.6 New

- The input as a platform type with provenance (`cost-model-study#CP1`: `client_dataset`, `stated`, `public_source`) and low / base / high with a guarantor on every input (`#CR2`); the input table as the stage *Předpoklady*. Board M2 § 01b–03d (session B) shows it for ChemVitro: T9 with a filter by state, the decisive rows marked; the uncertainty type, the meaning of the ends and the reason for each decisive input; the estimate from the 36 plants; the scope check as a table of excluded items with where, by what document, price and months, grade; and the correlations as judgement with the stability test and the rule's re-application in three settings written down on 5. 11. 2026, before the computation.
- Distributions and the correlation record: PERT, declared rank correlations through a Gaussian copula with a validated rank-to-latent conversion and a valid latent matrix (the methodologist's item; the contract recommends it over sample-wide Iman–Conover reordering so the numerical error contract matches the sampler), a non-PSD matrix refused (`#CR3`); empirical joint draws, one row of a client table per future carrying several inputs together, the table pinned by SHA-256 (`#CR3b`, **new**); yes/no events with a probability and fixed and uniform values as T9 names them; people who disagree a mixture (`#CR3` in the pivot branch's copy of the plan, `chore/pivot-business-decisions` @ `d8dc8ec5`, as the scope cites it: "each input's distribution by its source, a client-data estimate its own distribution, a person's range PERT, people who disagree a mixture (ADR 0046 W14)"; board M2 § 02 for Velis; board MV5 lists it among what the methodologist signs for the Monte Carlo entry).
- The estimate from comparables as a Metodika method (`client-data-in-analysis#12`; the exemplar's third assumption).
- The scope check: what an estimate's document excludes against what each option's documents require, each exclusion an input with its range (`cost-model-study#CR1b`, **new**).
- The stability test declared before computing, as the futures map's chunk defines it (`sociomapa-systemu#SYS9b`, chapter 9).
- The state column and its three values, read from steps 2 and 3 (`deep-research-v2#DRV1`'s state for public inputs).

### 6.7 ChemVitro acceptance

The answer key's three traps live in this step. Each is a cost the brochure hides and a site document reveals; the study must find and price it.

| Trap | Where it is | What the exemplar priced |
|---|---|---|
| Wastewater, CZ2 Lipová and DE2 Lindenau | `vyjadreni_cov.pdf` p. 2 and 5: the sewer bylaw's sulfate and COD limits against the process effluent (1 800 and 2 400 mg/l); Lindenau's plant has spare capacity for 250 m³ a day against 430 needed | pre-treatment 6,7–14,1 mil. EUR at both, from 8 comparable plants; Lindenau's direct-discharge permit 0–9 months (T9) |

| Trap | Where it is | What the exemplar priced |
|---|---|---|
| Air, CZ1 Vrbice and PL2 Bory | `ovzdusi_stanice.csv` 2025 values over the limits, combined with `memo_povolovani_ovzdusi.pdf` p. 2–3: compensation measures and 9–15 / 12–18 months longer permitting | compensation with probability 0,8, 3–15 mil. EUR, 9–18 months (T9); both sites then miss the deadline in most futures (T1) |
| Logistics, PL1 Odrzyca and DE3 Eschfeld | `vyjadreni_doprava.pdf` p. 7: the investor builds the 3,4 km spur and buys the land; p. 6: the bridge carries 16 t, tankers weigh 40 t | the spur at 1,9–2,8 mil. EUR per km plus road haulage until built; the bridge 3,2–5,0 mil. EUR and 12–30 months to heavy-traffic approval (T9) |

- Overrun from the comparables, median 17,0 %, not the estimator's band; the duplicate removed first (answer key § 3).
- The decoys left alone: Kamenec's single 2021 exceedance (under the target since); Eschfeld's plant charges from COD 2 500 mg/l against 2 400 in the effluent; Lipová's turning lane at about 2 mil. Kč; Bory's municipal plant does accept the effluent.
- Wendelstedt's park fee of about 1,1 mil. EUR a year, named only in the wastewater statement (p. 4).
- Criteria: K2, uncertainty fitness, is this step's criterion: decisive distributions, dependence and stress cases justified. K1's second question (material costs and dependencies in the model) is answered by the scope check.

CHAPTER 7 · STEP 5

# *Model*: one computation for every option

The model computes the measure the rule reads for every option in the same futures. It is code, pinned by version and seed, and a rerun gives the same bytes. Nothing in it is a forecast or a legal opinion.

## 7.1 What goes in

- The input table of step 4, frozen as a revision; the options and the baseline of step 1; the rule's measure.
- The timeline every option shares: for ChemVitro, decision and permit applications on 1. 2. 2027, permitting 8–11 months plus the site's extensions, construction 22 months plus the drawn comparable plant's delay, ramp-up at 60–90 % for six months (Appendix B).
- The seed and the number of futures, or a stopping rule that sets the number.

## 7.2 What comes out

- **The base case:** one computation with every input at its likely value. It compares the options and feeds the sensitivity analyses; it is not an estimate of what the client will save (R2-P10).
- **The futures:** N seeded draws of all inputs at once, with their correlations, each future computing the baseline and every option. From them: the median, the 90 % interval, and every probability as a share of futures (R2-P04 § 3.3).
- **The extra computations** on the same model and the same futures, each changing one assumption: the decision boundary over an assumption's range, delay of the decision, hiring delay, alternative terms, correlation settings (Appendix B).
- **The run record:** model type and version, input revision, seed, N, the methods pinned, the artefacts with their SHA-256 (MV0 M-2, M-3).

## 7.3 Checks

- Every option is computed in the same futures by the same model; the baseline is one of them (R2-P04 § 3.3).
- The same inputs and seed give the same bytes (M-3). This verifies the implementation, not the model's fitness for the decision (review R1; chapter 14).
- The number of futures follows the convergence rule of chapter 2: 20 000 independent joint futures, doubled at declared checkpoints up to 1,28 million, until every gate and decisive comparison clears a repeated-look-valid exact binomial bound; a comparison that has not cleared prints as unresolved. A fixed N and a ±0,5 pp margin alone certify nothing; the exemplar's N = 20 000 (R2-P04 § 3.3) is the starting point, not the criterion.
- Cash flows are discounted at the rule's rate, nominal or real as declared, with the convention named (mid-year, inflation 2 % provisional; Appendix B).
- Tax is inside the model as a fixed parameter per country, so the after-tax view agrees with every other view (review I4). The rule reads the pre-tax figure.
- What is outside the model is listed where the model is described, not discovered in the limits (R2-P15: transfer prices, hiring time in the schedule, raw-material freight, the annual step).
- An extra computation changes one assumption; combinations that are not computed are said not to be (R2-P15).

## 7.4 How the exemplar did it

The cost of supply is the investment, operations from start-up, VTX-40 bought until start-up, purchases above 40 000 t, less a residual value of 30 % of the investment in 2044; the saving of a site is the baseline's cost less the site's, in NPV, positive when the plant is cheaper (R2-P04 § 3.3). Purchases follow the supplier's formula to 30. 6. 2030, then the renewal premium, or the spot price while the plant is late (Appendix B). Operations are six items per tonne: raw materials at 540 EUR/t delivered, electricity and gas at the sites' offers moving with the country's price path, $CO_2$ at the allowance price less free allocation, water, sewer, hazardous waste and product freight at the quotes, 95 staff at the operator wage times a wage mix and contributions, maintenance and insurance at 2–4 % of the EPC estimate (Appendix B). The futures number 20 000 with seed 20270201. The base case carries the median overrun of 17 %; the futures draw one comparable plant for overrun and delay together. Eight extra computations reuse the same futures, each changing one assumption (Appendix B).

### 7.5 In AIA today

- Runs as workflows on the worker, with lease-fenced steps, artefacts, provenance and dependency edges, each artefact pinned by SHA-256 (`apps/worker`, `infrastructure/artifact_repository.py`, `domain/workflow.py`, `domain/pipeline.py` fingerprints).
- A run pins the organization's methodology in force and records it in its identity (`domain/methodology/pins.py`, `application/research.py`).
- The deterministic simulation core: a frozen WorldModel, reject-not-clip validation, seeded scenarios and frozen results (`domain/simulation/`). Its discipline is the model's; its content is not.
- Seeded synthetic data from `random.Random(seed)` and the Sociomap golden fixtures pinned by SHA-256: the rerun-equals-bytes practice (`packages/aia_core/tests/fixtures/sociomap/`).
- Budgets and cost ceilings per run, which a model computed by code does not need (a CALCULATED run has a known 0,00 USD ceiling, `cost-model-study` CP5).

### 7.6 New

- The platform: the model-type contract and the determinism harness (`cost-model-study#CP2`), the kind and the working copy (`#CP3`), the run with inputs frozen by artefact id (`#CP5`), the API (`#CP6`), the ESTIMATED grade with seed and method pinned and the Metodika module of defaults (`#CP10`). Board M3 (session C) shows the type for ChemVitro beside Velis: the shared timeline as one drawing (§ 02) and the eleven items of the cost of supply with their kind and base formula (§ 05), every number held to the exemplar by `tools/results_build.py`.
- **A model type of its own, `site_cost_of_supply`** (`cost-model-study#CM8`, **new**): a timeline per option whose dates are inputs, purchases, operations, residual value, tax as a fixed parameter, yearly cash flows, NPV before and after tax at a stated rate, T4–T6 as its results, and the convergence rule of chapter 2 (the futures double until each decisive comparison resolves, or print as unresolved). The planned type *scénáře nákladů* (`#CM1`–`#CM7`) has no timeline and no bought baseline, so the scope makes type 2 a second type rather than an extension.
- Monte Carlo with the input table's distributions and correlations (`#CR3`), the joint draw from comparables (`#CR3b`), PCG64 under the run's seed; board M3.
- The cash-flow schedule, NPV and discounted payback, and IRR by a stated root rule (`#CM7`; `quantitative-families#FIN1`, `#FIN2`).
- The extra computations as declared operations on a stored run, each one change at a time and said so: the boundary, delay and hiring tables, post-2035 and correlation recomputations (`cost-model-study#CR7b`, **new**, chapter 8; board M4).

### 7.7 ChemVitro acceptance

- Before ChemVitro, the oracle fixture O-CF (chapter 14): two years, 10 % discount, present costs 173,554 / 154,132 / 158,843 and a 12,1 receipt in year 2 worth exactly 10. Then seed 20270201 and 20 000 futures as the first checkpoint; the same seed gives the same result (`STUDIES.md` § 2, MV0 M-3).
- The spot price for a late start: any VTX-40 needed after 30. 6. 2030 and before the plant runs is bought at spot, as in February 2026 (`CASE.md` § 8).
- Tax reliefs out of the pre-tax figure the rule reads; after-tax at each country's rate (answer key § 3).
- Every option on the same timeline from 1. 2. 2027; a site's extra months (compensation, the bridge, the spur, the permit) enter its own schedule (R1 question 1).
- Criteria: K1's first and second questions (comparable options; material costs and dependencies); K4, a revised input invalidates the run's results and approvals.

CHAPTER 8 · STEP 6

# *Výsledky*: the rule applied, and what could change it

The results answer four questions in order: what the rule chooses, how far the verdict is from flipping, what information would change it, and what the finance, operations and schedule of each option look like. The first question is the recommendation; the rest is evidence about it.

## 8.1 What goes in

The futures and the base case of step 5; the rule of step 1; the extra computations declared in step 5.

## 8.2 What comes out

Five views, each a set of tables the report reuses (the exemplar's T1–T6, R2-P06 to R2-P11):

| View | Content | Exemplar |
|---|---|---|
| *Rozhodnutí* | the rule table per option (the measure's median and 90 % interval, each condition's value against its threshold, pass or fail), at the rule's rate and the client's other rates; the **decision boundary**: the rule's choice as one decisive assumption moves over its range; delay tables; stress tests; the tornado; break-even values; alternative terms; the correlation check | T1–T3; the flip at 14 / 15 / 16 %; delay 0/2/4/6 months with and without compensation; hiring delay; tornado; break-even prices; post-2035 terms; three correlation settings (R2-P06–P08); board M4 § 02–04 |
| *Hodnota informace* | two measures, in this order: **the value for the rule** (what knowing an answer in advance and applying the rule again would bring, averaged over the possible answers), then **EVPPI** (the gain in mean saving from knowing one input); then **the enquiry table**: what would be learned, the output, the duration, the cost, and the consequence of waiting | 122,5 mil. EUR for the rule at Odrzyca against 4,9 by EVPPI; four enquiries (R2-P09); board M4 |
| *Finance* | pre- and post-tax NPV at the rates, IRR, payback, investment by item with the probability of staying under the ceiling; incentives by their nature | T4, T5 with P(investment ≤ 175) and its percentiles (R2-P10); board M4b § 02–03 |
| *Provoz* | cost per unit by category in a full year of operation, base case | T6, 2032 (R2-P11); board M4d § 04, each column named as its leaf; freight, the spur, the bridge and the trucking surcharge on board M4f § 02–05 |
| *Harmonogram* | the start-date distribution per option against the deadline (median, 10th–90th percentile), P(on time), and the effect of each month of delay | the start chart and the delay tables (R2-P07); board M4g (new, session C): the start against the deadline per site, what lengthens each tail, and decision delay × hiring delay as one grid whose empty cells the report did not compute |

## 8.3 Checks

- The rule is applied exactly as written, condition by condition; the pass column is computed, never typed. The report prints the value of every condition against its threshold, pass and fail alike (T1).
- The verdict is given as a boundary, not a point: for the one or two assumptions the verdict rests on, the table of what the rule chooses at each value (R2-P06 § 5.3), each cell carrying its resolution state under the convergence rule. The exemplar's 15 % column reads Kamenec 0,804 and Odrzyca 0,797 against 0,80 on 20 000 draws; the 95 % interval of either covers 0,80, so that column is unresolved until the draws double enough to clear it, and the report says so instead of "at 15 % Kamenec". A verdict with no boundary table has not been tested.
- A probability is a share of the model's futures under the inputs that produced it; every probability says which assumption it stands on (R2-P04 § 3.3). The report claims no probability for anything the model does not compute (R2-P15).
- The value of information is computed for the rule first and for the mean second, through declared answer branches: each enquiry states its possible answers, their reliability, cost, duration and the inputs it shares with others; the joint model is conditioned on the answer, the timetable shifts, the cost of asking enters, and the rule is reapplied (chapter 2). Neither measure is a budget for the enquiry (review R7; K5). The rule's value and the expected-benefit EVPPI are different numbers and the report names which it prints.
- A median is not a mean: the rule's choice of statistic is kept, and the report says what the right tail does to the other (R2-P06 § 5.1).

- The base case and the futures may disagree; the report explains why when they do (R2-P08 § 7.1: break-even in the base case against loss in the median).
- Every extra computation names the one assumption it changes (R2-P15).

### 8.4 How the exemplar did it

No site passes at 8, 6 or 10 % (T1–T3). The Czech and Polish sites fail only the deadline condition (0,75–0,76 against 0,80 for the three leaders, 0,19–0,21 for Vrbice and Bory); the German sites fail mainly the loss condition (0,14–0,21 against 0,10). The verdict is a boundary: at a compensation chance up to 14 % the rule picks Odrzyca, at 15 % Kamenec, from 16 % nothing; the model uses 20 % (R2-P06 § 5.3). Delay of the decision by 2, 4 or 6 months lowers every P(on time); with compensation ruled out, the rule still picks Odrzyca after two months and nothing after four (R2-P07 § 6.1). Hiring delay is a stress test that adds to decision delay (§ 6.2). The tornado puts the 2026 Hanshin price first (a swing of 134,6–152,7 mil. EUR), the renewal premium second (57,7) and overrun third (34,2–44,5) (R2-P08). Break-even prices lie 1 249–1 308 EUR/t for Czech and Polish sites, 1 413–1 450 for German ones; at the 2025 average of 1 296 EUR/t the median of every site turns negative (§ 7.1). Post-2035 terms are the second lever: at a 30 % premium Wendelstedt would pass (§ 7.2). Three correlation settings move medians by at most 1,4 mil. EUR and probabilities by at most 0,01 (§ 7.3). The value for the rule of the authorities' answer is 115,2–152,1 mil. EUR; EVPPI of the same information is 4,9 mil. EUR, of all inputs 14,2 (R2-P09). Finance: base-case saving 170,7 / 165,9 / 159,9 mil. EUR for the three leaders at 8 %, IRR 18,2–20,5 % in Czechia and Poland, payback 2034–2036; after tax Kamenec and Odrzyca are level (T4). Operations: 1 290–1 311 EUR/t in Czechia and Poland against 1 434–1 463 in Germany in 2032, the difference in staff, maintenance and freight (T6).

### 8.5 In AIA today

- A number reaches a result only as an admitted claim (`domain/evidence/admission.py`, `application/admitted_results.py`); suppression is decided on the server and a suppressed row prints its reasons, never a number. The pattern holds for model results: a cell the method does not support is not printed.
- The results screens of a research run and their reading in words (`ExecutionSteps.tsx`, `lib/result-reading.ts`): unknown is never ordered, an interval missing means no order.
- The report's print grades and the evidence ledger (`domain/report/evidence.py`), the per-question tables as exports (`domain/report/table_export.py`, RD12).
- The budget line and spend reading as the pattern for "unknown never zero" (`lib/budget-line.ts`, `lib/run-spend.ts`).

### 8.6 New

- The rule judged as written (`cost-model-study#CR7`) and, new, the rule at three rates, the decision boundary over one input's range, delay and hiring tables, post-2035 and correlation recomputations, each one change at a time and said so (`#CR7b`, **new**); the method on screen (`#CR8`); grades ESTIMATED for everything from the futures and CALCULATED for the base case (`#CP8`, `#CP10`). Boards M4, M4b, M4d, M4f and M4g (session C) carry the ChemVitro tables beside the Velis sections, each result section naming the perspective leaf it feeds (chapter 12).
- Value of information for the rule: the rule re-applied per declared answer branch, the average of what it then picks against today's "nic", the enquiry's cost and duration inside the comparison, and the enquiry table with its effect on the deadline (`#CR6b`, **new**; K5); EVPPI beside it (`#CR6`), its estimator's bias and precision checked on the oracle fixture O-INFO (chapter 14).
- Break-even and one-way sensitivity (`#CM2`), global sensitivity beside the tornado (`#CR4`), three chart kinds the exemplar needs and the model lacks: the absolute stacked bar, the tornado, the line with marked points (`#CP8`).
- Finance: scenarios on NPV and IRR (`quantitative-families#FIN3`), tax relief against grant as legal parameters (`legal-context#LGL3`); board M4b.
- Operations: cost per unit by category in a stated year (board M4d); *Harmonogram*, the start-date distribution per option against the deadline, P(*včas*) and the months-of-delay table (`cost-model-study#CM9`, **new**; board M4g, `KM12Schedule.dc.html`, new).
- Boards M4c *Účetnictví* and M4e *Trh* are parked for type 2: they have no leaf in the ChemVitro case. The Výsledky row now runs M4d → M4g → M4f → M4e.

### 8.7 ChemVitro acceptance

The answer key's reference model finds CZ3 Kamenec passing at 8 % (median +39,8 mil. EUR, P(beats) 0,82, P(loss > 20) 0,09, P(start) 1,00), CZ2 close behind, PL1 on the edge of the loss limit, nothing at 10 %, and names the renewal premium (12–22 %) as the assumption most worth checking (answer key § 1). The exemplar finds no site passing at any rate and names the compensation chance first and the Hanshin price second. The two differ on one judgement: the reference prices offsets only where a station exceeds the limits; the exemplar adds a 0,2 chance everywhere else in Czechia and Poland, and says that this judgement decides the verdict and has no document behind it (R2-P15).

- The acceptance therefore does not require the exemplar's verdict or the reference's numbers. The key says so itself: "judge a report by whether it found the planted facts and applied the rule honestly, not by whether it hit these numbers." Since version 4 both readings are frozen as separate configurations of the oracle fixture O-CHEM (chapter 14): the reference's Kamenec 39,8 and the exemplar's 137,1 are each a target for its own configuration and never for the other.
- It requires: the rule applied as written with every condition's value printed; the verdict given with its boundary on the assumption it rests on; the assumption most worth checking named with its value for the rule; the three traps and the overrun from the comparables visible in the investment and the schedule (CZ1 and PL2 fail the start date; DE3's P(start) falls with the bridge).
- Both the exemplar and the reference agree that the brochure favourite, PL2 Bory, cannot start on time. A study that recommends Bory has read the brochure, not the files.
- Criteria: K3 (the rule's choice kept apart from supplementary scoring), K5 (every enquiry states its effect on the deadline), K6 (the rule table, the summary and the boundary table say the same numbers).

CHAPTER 9 · STEP 7

# *Sociomapa*: the futures map

A Sociomap places the options as points in a landscape: two options lie close when the same observers rate them alike, and the height shows the mean rating. In a type-2 study the observers are the model's futures. The map is evidence about the structure of the options' risks, never about the recommendation.

## 9.1 What goes in

- The futures of step 5 as a table: one row per future, one column per option, the cell the measure the rule reads (the saving against the baseline in that future).
- The mode contract for futures (board MV3 § 02, mode *Scénáře*): what a row is, what the claim is about, what is measured, what the null means, the resampling, what the stability measures.
- The stability test, declared before the computation: which alternative settings the map is recomputed under and what agreement counts as stable.
- The method pinned: `aia-sociomap-5`, the organization's methodology in force.

## 9.2 What comes out

- **The futures map** in two views: the 3D terrain and the top view that does not distort distances by perspective (R2-P12 § 11.2). Heights are mean saving; links are drawn only between pairs the method marks reliable; a warm link means co-movement, its width the relation's strength.
- **The relation matrix**, the exemplar's T7: each pair's relation across the futures, from which the map is drawn.
- **The shared-failure table**, T8: when an option is worse than the baseline, the share of those futures in which each other option is worse too, printed with its denominators (how many bad futures the share is of). This reads the absolute, uncentred futures; the map reads the centred relations, while its heights are the uncentred mean savings.
- **Three separate results, never one word:** layout convergence (did the layout algorithm converge in every setting), representation fit (how well the layout represents the relation matrix) and robustness (the agreement of the distance orders across the declared assumption settings against the threshold). A grouping the report keeps after a failed robustness test is tested on its own, or the matrix is printed without a spatial reading.
- **The register status:** the map's method at its spec fingerprint, approved or *interní* (`domain/sociomap_register.py`), printed on the map and in the report.

## 9.3 Checks

- **Mode contract.** A row is one independent draw of the model. The claim is about the model and its assumptions, not about the market. Centring each future on the mean of the options makes the map show relative performance: who does better in the same future, not whether all do well (R2-P04 § 3.3; review R5).
- **Relative apart from absolute.** The question "is option B a backup for A" is answered by the uncentred shared-failure table, not by the map: centring removes the common level, which is exactly what shared failure is made of (R2-P12 § 11.3; board MV3 § 05). The table answers downside dependence only; whether a site could operationally stand in for another (switching time, cost, feasibility) is a separate assessment the report must name as such.
- **Links only where the contract allows.** The exemplar drew links where the `-5` method's pair test, with the false-discovery correction, marked a pair reliable (2 of 28). Board MV3's contract for the futures mode (session D, § 02) says a null makes no sense when the model creates the relations, asks for no p-value, and measures stability as the sensitivity of the shape to assumptions declared in advance; its § 03 still draws the exemplar's two reliable pairs. Which of the two applies to futures is an open question for the methodologist (chapter 14); until it is answered, the report prints which rule it drew by.
- **Determinedness.** Split-half agreement of two halves of the futures against the method's threshold says the layout is not an artefact of the draw (0,91 against 0,59 in the exemplar). It does not say the layout is stable under the assumptions.
- **Stability, declared in advance.** The map is recomputed under the alternative settings written down in step 4 (the exemplar: no correlations, the judgement's correlations, strong correlations) and counts as stable only if every pair of versions agrees in the order of the inter-option distances at or above the threshold (0,8 in the exemplar, Appendix B). Failure is printed, not softened.
- **What the map never claims** (board MV3 § 06): causation; the behaviour of the market or the firm; that a relation is observed rather than a property of the model; that a distance is an effect size in any unit; that the terrain between points is an estimate; validity borrowed from team-communication Sociomaps, which do not cover this use.

- **Internal until approved.** Maps whose observers are futures or properties are an estimate awaiting the methodologist's approval; they print *odhad (metoda čeká na schválení metodikem)* and carry the internal label (R2-P04 § 3.2). The Sociomap of people's ratings is approved since 21. 10. 2026; this is a different mode.

### 9.4 How the exemplar did it

Eight sites, 20 000 futures as observers, each future centred on the mean of the eight (R2-P04 § 3.3). The map divides the sites by country: the German sites lie together in a valley (74,2 mil. EUR mean saving at Eschfeld), the Czech and Polish sites on a ridge (147,6 at Odrzyca). Within countries the relations are strong (Lindenau–Eschfeld 0,71, Wendelstedt–Eschfeld 0,70; Lipová–Kamenec 0,55; Odrzyca–Bory 0,48); between countries weak and negative, −0,07 to −0,27, the sign partly a consequence of centring (T7). Two of 28 pairs were marked reliable. Split-half agreement 0,91 against a threshold of 0,59: determined. The stability test failed: the judgement version agrees with the no-correlation version at 0,95 and with the strong version at 0,84, but those two agree only at 0,75, under 0,8; the relations in the matrix are steadier than the drawn positions (0,966, 0,899, 0,826). Without correlations seven pairs are reliable and the Czech sites merge; with strong correlations no pair is and Vrbice moves away. The report therefore reads from the map only the heights and the split into a German group and a Czech–Polish side, which appears in every version, and nothing about fine positions, causes or probabilities (R2-P12 § 11.2). Version 4 tightens that: the country split survived a failed test only by inspection, so the method requires the retained grouping to be tested on its own, and the version without correlations, which did not converge in the program's log (R2-P15), may not count as one of the three settings until it does. For the backup question the exemplar uses T8: the best backup fails together with the chosen site in 75–79 % of its bad futures, the German sites in 97–100 % (§ 11.1, 11.3). Its sentence "no site is a backup of another" is a claim about tested downside dependence for the two focal sites; the method prints it with that scope.

### 9.5 In AIA today

- The `aia-sociomap-5` engine, signed off by the methodologist on 21. 10. 2026 and what a new run pins: `domain/sociomap/signed_off_inputs.py` (per-person centring, the baseline, pair floors, Benjamini–Hochberg), `signed_off_layout.py`, `signed_off_credibility.py` (split-half agreement, map uncertainty), `signed_off_respondents.py`, `signed_off_segments.py`; `engine_v2.py` `compute_object_map` and the contract-3 artefact.
- The register: which method at which spec fingerprint is approved; anything absent is `INTERNAL_ONLY`, never presumed (`domain/sociomap_register.py`, ADR 0038).
- Map snapshots drawn on the server from the stored map at the report's camera and approved as artefacts (`application/map_snapshots.py`, RD4); the map chapters of the report (`application/report_maps.py`, `infrastructure/report_docx/object_map_figure.py`).
- The canvas's own proof that the engine takes a futures table as rows: `prototypes/obchodni-model/tools/engine_maps.py` ran `compute_object_map` on 2 000 futures and drew the maps with the report's renderer; session D's `build_maps.py` and `engine_weights.py` did the same for the exemplar's futures map in its three settings and for 1 806 tested weightings.
- The web's map: `ObjectMapPanel`, `lib/object-map-signed-off.ts`, `lib/sociomap-display.ts`, display only.

### 9.6 New

- The futures mode's contract (`sociomapa-systemu#SYS9`): rows a model's draws, each centred on the mean of the objects; the claim about the model and its assumptions, not the market or causes; no p-value; determinedness from the maps of two halves of the futures (threshold 0,59); stability as the shape's sensitivity to assumptions declared before the computation. Board MV3 (session D) prints it as the first of five readings of one engine (futures, experts, units, periods, and the weighting map as the fifth), and renders the exemplar's map in § 03–07: the map in the judgement's setting, T7, T8, the relative and absolute readings side by side (Odrzyca · Kamenec −0,14 on the map against 75–79 % shared failures; Odrzyca · Eschfeld −0,15 against 99 %), and the stability test with its three results (0,95 · 0,84 · 0,75) and verdict.
- The shared-failure table T8 and the stability test: Spearman on the 28 distances between every two settings ≥ 0,8, three settings declared before computing, a failure printed as the exemplar prints it (`#SYS9b`, **new**; board MV3 § 05 and § 07). For the Velis case the board says the three-setting test is not yet written and must be declared before the next computation.
- Each mode its own Metodika entry, internal until approved (`#SYS12`); the map on Výsledky and in the report (`cost-model-study#CS5`), declared extremes and direction per property (`#CS1`, reused by the weighting map).
- The weighting-scenario map, a second map beside this one, on board MV9 (`sociomapa-systemu#SYS13`, **new**; chapter 12). SI1–SI4 keep only what MV9 reads: states, gates, coverage, the overlap check.

### 9.7 ChemVitro acceptance

- The eight sites placed by whether they do well in the same futures; links only where the declared rule allows; the view chosen and the reason stated (`STUDIES.md` § 2).
- The map labelled internal until Metodika approves the method.

- The relation matrix and the shared-failure table printed; the backup question answered from the table.
- The stability test declared before the computation, and its result printed whether it passed or failed.
- Criteria: K3 (the map is not part of the recommendation), K6 (the matrix, the text and the figure agree).

CHAPTER 10 · STEPS 8 AND 9

# *Osnova a Zpráva*: the outline, then the report

The outline comes before the text. It lists every claim the report will make in the order of the argument, each with its source or method and why it matters to the decision. The report then says only what the outline holds.

## 10.1 What goes in

- The results, the map, the input table, the findings table, the public inputs and the decision record: every artefact of steps 1–7, by id.
- The part structure of a type-2 report and the grade and source conventions (below).
- The validity review of chapter 14, before approval.

## 10.2 What comes out

**The outline (*Osnova zprávy*, board MV4)**

Five parts in the order of the argument (review R10): 1 the decision and what is asked of the board; 2 why the preferred option is acceptable, or why none is; 3 what could change that conclusion; 4 implementation conditions, owners and dates; 5 supporting evidence and technical appendices. Each point carries its source or method, its grade attributes, and one line "why this matters" (*proč na tom záleží*). A judgement is a typed judgement record; a synthesis from sources is an inference record with its evidence (review I5). Stops (*zastávky*) close parts for review; a closed stop may go to the client's decision-makers for comment. Board MV4 (session E) shows the outline of report 2 as 18 parts with their ids, maps the five parts of the argument onto them (1 = the summary, P01; 2 = chapters 5–6, P06–P07; 3 = chapters 7–8 and 12, P08–P09 and P13, where the weighting-scenario map belongs; 4 = chapter 13, P14; 5 = the rest), prints the seven grades with their four attributes, and opens a number's provenance card: source, method, uncertainty, validation, the run, the inputs it stands on, the data check behind it.

**The report (*Zpráva*, board M6)**

The exemplar's part structure, which the MVP reproduces:

| Part | Content | Rule |
|---|---|---|
| R2-P01 | *Shrnutí*: *Doporučení* · *Důvod* · *Jistota* · *Další rozhodnutí*, the rule table, *Stav podkladů* | one page plus the state box; the rule table is T1 without the intervals; the state box names the gaps that could change the result and the public inputs not yet verified |
| R2-P02 | *Úvod*: the situation, the options, the rule | the rule printed in full with its date and the interpretation record |
| R2-P03 | *Přehled literatury* | each work with "what follows for this report" |
| R2-P04 | *Metodika*: about AIA, the Sociomap, the model in words, the data, the grades | the grade legend is printed in every report; the Sociomap paragraph says which modes are approved |
| R2-P05 | *Předpoklady*: the assumptions the verdict rests on | the ones without a document first |
| R2-P06–P11 | the result chapters: decision, schedule, levers, information, finance, operations | each chapter's first sentence is its finding; each number carries its grade; sources numbered at the chapter's end |
| R2-P12 | the map chapter | relative and absolute readings, the stability result, what the map may and may not say |
| R2-P13 | *Diskuse* | what the results mean together, and what the model does not resolve |
| R2-P14 | *Doporučení a provedení* | a dated plan: when, what, who, output; the enquiries with their owners; what to watch |
| R2-P15 | *Meze studie* | every judgement without a document, every provisional input, every simplification, every method awaiting approval; "each change computed separately" |
| R2-P16–P17 | *Literatura, Seznam zkratek* | — |
| R2-P18 | *Přílohy* A–D and tables T1–T9 | A the document register; B the method in detail and the findings P1; C data checks and computation (seed, N); D the tables, generated by the model from the same run |

Every number carries one of the grades of chapter 13 in small type and a numbered source at the chapter's end; the summary's recommendation status says *předběžné* where the recommendation rests on a method outside the register (chapter 13). Board M6 (session E) renders the exemplar's page 1 and chapter 13 as AIA would print them and adds the release state: *Předběžné doporučení*, four unresolved qualifications (sixteen public inputs unverified, the compensation chance a judgement without a document, the investment reading awaiting the CFO, the futures map failed its stability test), one K6 finding, and *Schválit jako předběžné* as the only approval offered. Its summary gains a paragraph the exemplar lacks, *Jiné vážení* beside *Jistota*: which sites win under the same weights of the perspectives, as a share of tested settings, never a probability, and not part of the recommendation; the map itself sits in the report's chapter 12 (P13), inside part 3 of the argument.

### 10.3 Checks

- No point without a source or a method; a point whose source was not captured does not enter the report (MV0 M-9; board MV4's point 3.4).
- The report says only what the closed outline holds. A sentence that compares in words with no number behind it is marked (RD7's sentence check, already built).
- Every number is a reference to a stored result: the same run, the same artefact id. Tables are generated by the model from the run, so they agree with the text by construction (R2-P18 Appendix D; K6).
- The required disclosures are in every report: the recommendation's status and qualifications, each number's source and method, what is outside the model, who stands behind the assumptions, the method versions (board MV4's *Povinná zveřejnění*).
- The validity review is recorded before approval, and the approval records the exact version and the unresolved qualifications (review R1, R9; chapter 14).
- A stop closed with the client's people records their agreement (ADR 0046 W26).
- No respondent, person or e-mail is named beyond a role; the report is Czech in the MVP (ADR 0046 W28).

### 10.4 How the exemplar did it

The report's first page is the summary in four paragraphs and the rule table; the second is the state of the inputs. Every chapter's title is its finding ("No site yet passes the rule", "The cheapest sites do not meet the deadline certainty", "The saving rests on today's price from Hanshin"). Every number carries its grade in small type, and every chapter ends with *Zdroje a stupně*, numbered. The part ids R2-P01 to R2-P18 let the executive comment part by part. The limits chapter admits what the report cannot claim: the compensation chance is a judgement, public inputs are provisional, the map is not stable, the methods of the futures await approval, each change was computed separately (R2-P15). The implementation plan is a dated table with owners and outputs (R2-P14). The tables T1–T9 are generated from the same run as the text (R2-P18).

### 10.5 In AIA today

- The report as data: the document model with sections and blocks, numbers only as refs (`domain/report/model.py`, `document_model.py`); the evidence ledger built from admitted claims with print grades (`evidence.py`); the section-source manifest naming each section's sources (`sources.py`); printed numbers as locked content controls with one tag grammar (`control_tags.py`, `controls.py`).
- The DOCX renderer with deterministic bytes, the Word style sheet from the print tokens, tables with repeating headers and suppressed rows counted, evidence marks as vector images (`infrastructure/report_docx/`); the PDF beside it (`infrastructure/report_pdf.py`).
- Composition: the internal report, the client report, the report with literature, their chapters from verified sources (`application/report.py`, `report_chapters.py`, `report_tables.py`, `report_design.py`, `report_method.py`, `client_report.py`, `contextual_report.py`).
- Approval once per version with the approver recorded and self-approval rules (`application/report_approval.py`, `artifact_repository.approval_record`); the client gate's checks as warnings (`domain/report/client_gate.py`, RD0); the sentence check (`sentence_check.py`, RD7); the report editor with saved versions and their changes against what code wrote (`report_editor*.py`, `saved_changes.py`).
- The stored document read on screen with each number a chip to its claim (`ReportDocumentView`, `lib/workbench-artifacts/report.ts`).

### 10.6 New

- The outline as data with a resolving link on every point (`research-findings-stage#FND0`, `#FND2`), findings in (`#FND3`), tables from points and graphs from tables (`#FND4`, `#FND5`), stops (`#FND8`), *Zpráva* written from the outline (`#FND9`); board MV4 (session E, landed) is the outline of report 2 itself.
- Composition per situation from chapter kinds and outline presets (ADR 0045; `situation-reports#RC1`), fixed method texts signed by the methodology owner (`#RC4`); the report-2 recipe: R2-P01…P18 as an outline preset with part ids, the grade legend, numbered sources per chapter, *Stav podkladů* and *Literatura* (`research-findings-stage#FND13`, **new**); the argument order (review R10).
- The frame and CALCULATED (`cost-model-study#CP8`), ESTIMATED (`#CP10`), the method appendix (`#CR9`), and the seven print grades mapped to the code's with an admission rule each (`#CP8b`, **new**; chapter 13).
- Approve and export (`report-delivery#RD3`); tables generated from the run into the report (the exemplar's T1–T9, `#FND4`); the limits chapter composed from the input table's states and the run's declared simplifications (part of `#FND13`).

- The validity review before approval: the five questions, extreme-input tests and a named reviewer recorded on the ledger (`research-findings-stage#FND14`, **new**; board MV8, chapter 14); the release label of R9 where the owner decides it (board M6, *Stav vydání*).
- The state box and the limits chapter composed from the input table: board MV4's P01 takes its count from T9 (16 of 67) and names the text's 54 as the K6 finding; the *prozatímní* banner on every page while one decisive input is provisional (board MV2 § 02b).

### 10.7 ChemVitro acceptance

- Every number carries its grade and source: file and row or page, or the quoted public source (`CASE.md` § 11).
- An estimate from a method that is not approved is labelled *odhad (metoda čeká na schválení metodikem)*.
- The report is Czech, as PDF or DOCX; the seven parts of `STUDIES.md` "What every report contains" are present.
- No respondent named or quoted by id.
- Criteria: K6 is checked here, by code and by a person: no contradiction between narrative, tables and figures, percentile and probability wording included. The canvas's own K6 finding (SI3 against KM4) shows the check is needed.

CHAPTER 11

# Built vs designed

No type-2 code is on `develop`. The exemplar's model ran outside the product. What AIA has is the ground the pivot builds on: the shell, the working copy, client sources, Deep Research, the Sociomap engine and its register, the report document model and its renderer, approvals and audit. This chapter says which is which.

## 11.1 The state on 9 November 2026

| Step | Built on `develop` (reused) | Designed (board; plan chunk as the scope names it; **bold** = new, in no plan file yet) | Neither yet |
|---|---|---|---|
| *Zadání* | working copy, revisions, edit ops; Brief stage; the Lead; approval ledger; material classes | Z1–Z7; `decision-brief#DB0`, `DB2`–`DB5`, **DB3b**; `cost-model-study#CP3` (the model inside the one merged study: no new study kind, A2) | the rule freeze with amendments as revisions (chapter 2) |
| *Data* | client sources with readers and locators; dataset preview with personal columns; intake sheet; evidence policy (ADR 0027/0037) | M1, MV1; `client-data-in-analysis#2`, `2a`, `3`, `3a`, `3b`, `5`, `10`, `11`, `12`, **`13`**; `cost-model-study#CR1` | — |
| *Rešerše* | Deep Research with grounding and the sealed bundle; RetrievalGate; Wikipedia, Brave, Common Crawl; verified sources | MV2, MV2b; `deep-research-v2#DRV1`–`DRV3`, `DRV5`; `public-data-routes#PDR1`, `PDR2`; `legal-context#LGL1`–`LGL3` | verification at source as a dated act, named in `DRV1`'s state |
| *Předpoklady* | methodology as data (versions, activation, bounds); ledger and audit; the simulation core's typed, validated inputs as a pattern | M2; `cost-model-study#CP1`, `CR2`, `CR3` (mixtures included in the pivot branch's copy), **CR3b**, **CR1b**; `client-data-in-analysis#12` | — |
| *Model* | worker, workflows, artefacts with SHA-256, pipeline fingerprints; methodology pins on a run; seeded determinism | M3, M4; `cost-model-study#CP2`, `CP3`, `CP5`, `CP6`, `CM7`, **CM8**, `CR3`, `CP10`; `quantitative-families#FIN1`, `FIN2` | — |
| *Výsledky* | admitted claims and suppression; results reading (unknown never ordered); print grades; table exports | M4, M4b, M4d, new M4g; `cost-model-study#CR7`, **CR7b**, `CM2`, `CR4`, `CR6`, **CR6b**, `CR8`, `CP8`, **CM9**; `quantitative-families#FIN3`; `legal-context#LGL3` | — |
| *Sociomapa* | `aia-sociomap-5` engine; the register; map snapshots; report map chapters; the canvas tool that maps futures with the engine | MV3, SI1–SI4, new MV9; `sociomapa-systemu#SYS9`, **SYS9b**, `SYS12`, **SYS13**; `cost-model-study#CS1`, `CS5` | the register entries themselves (the methodologist's) |
| *Osnova, Zpráva* | document model, evidence ledger, control tags, source manifest; DOCX and PDF renderers; composition; approval; client gate; sentence check; report editor | MV4, M6, MV8, MV5; `research-findings-stage#FND0`, `FND2`–`FND5`, `FND8`, `FND9`, **FND13**, **FND14**; `situation-reports#RC1`, `RC4`; `cost-model-study#CP8`, **CP8b**, `CP10`, `CR9`; `report-delivery#RD3` | the release label of R9 (the owner's) |

## 11.2 The plans that carry the build

From `python tools/progress.py` on 9 November, on `develop`: `cost-model-study` is planned, 0 of 32 chunks; `client-data-in-analysis` in progress, 1 of 15 (8a built); `report-delivery` in progress, 7 of 16; `quantitative-families` and `situation-reports` planned. The other plans the scope cites (`decision-brief`, `deep-research-v2`, `public-data-routes`, `legal-context`, `sociomapa-systemu`, `research-findings-stage`, `analysis-frameworks`, `diagrams`, `register-personal-data`) exist only on branch `chore/pivot-business-decisions`, with ADR 0046, Proposed and not merged; the scope's preamble says so and reads them at `d8dc8ec5`. The chunks the scope marks **new** exist in no plan file at all. ADR 0044 (quantitative models as a study kind) and ADR 0045 (reports composed per situation) are Proposed on `develop`. The scope (§ 2, § 7) maps each step to its chunks and orders them in five waves; this chapter's "Designed" column is that mapping.

## 11.3 Three honest statements

- **The exemplar was not produced by AIA's product.** Its author computed the model with his own program and wrote the report by hand to AIA's conventions. The report's sentence "every number was computed by a verifiable program" is true of that program. The MVP's job is to make it true of AIA.
- **The canvas is design.** The thirty boards show the screens and the method; their numbers are fictional, and the Velis case's numbers are illustrative, not validation evidence (board MV5). The boards hold the method; the plan chunks hold the build.
- **The methods of the futures are not registered.** Monte Carlo with PERT and correlations, the decision boundary, both measures of the value of information, the futures map and the weighting-scenario map await the methodologist's entries (chapter 14). Until then their results print *odhad (metoda čeká na schválení metodikem)*, and a recommendation that rests on them is *předběžné* under R9, adopted by the owner on 9 November (chapter 2).

### 11.4 Not in this release

Cut from the ten weeks on 9 November, to keep the spine to what the exemplar needs: the framework catalogue and picker in *Zadání*; model types other than `site_cost_of_supply`; automatic public-data and legal connectors (public inputs stay typed and sourced, entered by hand); fitted demand forecasting; global sensitivity by Sobol or PRCC; the respondent, expert, unit and period map modes; survey processing; the generic diagram editor; the shared client-review interface; monitoring. Each stays in its plan file as parked.

CHAPTER 12

# Perspectives, the cost partition and the two maps

The owner asked for every business perspective: financial, strategic, operational, marketing, logistics, "just everything", with weights on them run as scenarios and shown on the Sociomap. Version 4 gives that request one construction: every pre-tax cash-flow line of an option is allocated to exactly one leaf under exactly one perspective, a perspective's total is the discounted cost of its leaves, and one value function and one weight vector score every option alike. Overlap is impossible by construction, and the owner's "logistics 0,3, operations 0,5" means exactly what it says.

## 12.1 The partition

A **leaf** is one cash-flow line (or a group of lines that always travel together), discounted at the rule's rate. A **perspective** is a set of leaves. The same expenditure never appears in two leaves; distinct expenditures caused by the same event may. Delay costs enter through the purchases they actually cause, not as a separate penalty. The allocation for a type-2 study, read from the exemplar's T5, T6 and its Appendix B:

| Perspective | Leaves (each a cash-flow line, discounted) | Source |
|---|---|---|
| *Finanční* | core plant capital; main-plot land; working capital where applicable; the remaining ordinary and spot purchases of VTX-40 until start-up; residual credits and other financial adjustments not allocated below | T5, Appendix B |
| *Strategická* | no scored leaf until an independently defined consequence is approved (open, R2-Q04). The shared-failure analysis of chapter 9 stays a table of its own, never a weighted leaf. | T8 |
| *Provozní* | raw materials; electricity and gas; maintenance and insurance, excluding the services allocated below | T6 |
| *Marketingová / obchodní* | no scored leaf for captive supply to Brno and Pardubice; the perspective is drawn and says why it is empty | R2-P02 |
| *Logistická* | product freight; spur construction and spur land; bridge works; temporary transhipment and road haulage until the spur is built | T5, T6, T9 |
| *Prostředí a povolování* | pre-treatment capital and operation; air abatement; compensation measures; water and sewer; hazardous waste; carbon (allowances less free allocation); separately identified environmental and park services | T5, T6, T9 |
| *Lidé* | wages, employer contributions and evidenced recruitment expenditure. Hiring time enters the timetable and therefore the gates, not a second weighted penalty. | T6, T9 |

Three checks hold the allocation: every expenditure the case documents is allocated once (the answer key's air abatement and spur land included); the perspective totals sum to the site's cost of supply; buying cost less site cost equals the NPV saving. A cost that cannot be interpreted comparably across sites (a one-off the documents give for one site only, with no counterpart elsewhere) is not admitted as a leaf; it keeps its range in the model and appears in the report as an exclusion.

Five rules carry over from the review of 5 November, two of them changed by this construction:

- **Gates are not leaves** (R2). The rule's four conditions are pass / fail / unknown per option, shown beside the map, never weighted.
- **A range stays a range** (R2). A leaf the files give as a range (the spur per kilometre, pre-treatment, the permitting extension) has a lower and an upper score bound, the direction applied correctly: for a cost the minimum raw value is the *favourable* end, so the map never draws it as the base. Both bound views are shown and labelled as bounds, not as jointly feasible futures. Nothing fills a missing fact: Bory's nearby hiring is 120 people because the client's file says so (`trh_prace_regiony.xlsx`, PL2), not a range.
- **One expenditure, one leaf** (R3). The partition replaces version 3's three roles (*úspora*, *v modelu*, *mimo model*) and its two scenario families. NPV saving, after-tax NPV, IRR and payback are printed beside the weighted score, unweighted, because they already contain every leaf.
- **Judgement is labelled** (A3). A leaf whose value is a person's judgement carries that person's name and the grade *úsudek*.
- **An empty perspective stays and says why**; a globally inactive perspective has weight zero for every site, approved as such. A nonzero weight on an empty perspective is refused with the reason; nothing is redistributed silently.

## 12.2 One scoring structure for every option

For a site $i$ and a perspective $p$, $C_{ip}$ is the discounted sum of $p$'s leaves. The approved value function $v_p$ maps it onto 0–10 on fixed anchors shared by all eight sites; a value outside the anchors triggers a review, never silent clipping. The weighted score is

$$S_i(w) = \Sigma_p \, w_p \cdot v_p(C_{ip})$$

- **One weight vector for every site.** No site loses a weight because a leaf does not apply to it: evidenced absence of an applicable expenditure is a cost of zero through the common scale; unknown expenditure is a range with two bounds. Version 3's rule that a null component "takes no weight there" is withdrawn: it compared sites under different trade-offs.
- **Leaves are an accounting breakdown, not equally weighted subcriteria.** Splitting a cost line (freight into rail and road) leaves the perspective total, the score and the rank unchanged. This is the admission test of every allocation, and version 3's "a perspective's weight shared equally by its admitted components" fails it.
- **Anchors and swing weights are approved together**, before any site score is shown (swing weighting, R3): a weight is a trade-off over a stated swing of a perspective's total, not a feeling about a category name. The additive form itself is reviewed with them.
- **A scenario is a named weight vector** the client's people approve; the researcher records it with its scales through the existing approval flow (the owner, 9 November, R2-Q06), with who composed it and when, before results.

The owner's example, with the remaining weight assigned to finance: *finance 0,20 · strategie 0 · provoz 0,50 · marketing 0 · logistika 0,30 · prostředí 0 · lidé 0*. "Finance" is its partition, the capital, land and purchases, not the total NPV. The NPV still includes every cost, including the perspectives given zero diagnostic weight. The scenario is admitted; version 3's refusal of "finance 0,5 · provoz 0,3 · logistika 0,2" was a refusal of the three-role mix, which no longer exists.

## 12.3 Ranges and sensitivity

A ranged leaf gives the site an interval of scores. A **robust leader** needs a validated positive lower bound on its score difference against every rival over all admitted unknowns jointly; overlapping intervals mean leadership is not established, and the map says so. For each named scenario the study tests, in this order and reported separately:

1. the approved centre;
2. Dirichlet fans at concentrations 20, 60 and 180, 300 settings each, drawn on the positive-weight coordinates only so an explicit zero stays zero;
3. directed weight transfers between pairs of perspectives, and a simplex grid where the number of active perspectives keeps it small;
4. the approved alternative anchors and the uncertain cost allocations;
5. the evidence bounds, coherent joint stress cases and simultaneous unknowns.

Results are shares of tested settings, per scenario, per fan and per evidence setting: the share in which an option is uniquely first, the share of ties and the share left unresolved by the bounds, which together reconcile to 100 %. Scenarios are never pooled into one popularity figure; version 3's "all tested settings: Odrzyca 35 %" is withdrawn with the families.

## 12.4 The Sociomap view of a scenario

Each scenario has its own view on the `aia-sociomap-5` engine, under the mode contract the methodologist registers for it (`sociomapa-systemu#SYS12`, `SYS13`):

- rows are the tested weight settings; objects are the eight sites; proximity is similarity of score response across the settings; height is the score under the selected setting;
- a ranged leaf keeps its paired bound views and score intervals in the table beside the map;
- the gate status of every site (pass / fail / unknown) is printed beside it; a hypothetical change of a gate's state is a named scenario, never an edit;
- the rule's verdict stands in a panel of its own; the map is preference information under *Co by závěr změnilo* (R3, R10) and never the recommendation;
- no respondent labels, no "reliable pairs", no respondent segments and no sampling-error circles appear: those belong to the respondent mode, whose statistics this mode does not read;
- the view records its layout fit, its numerical convergence and its sensitivity under the mode contract; geometry the contract does not support produces no spatial claim.

Board MV9 was built on 8 November under version 3's two families; its six scenarios, their leaders and its 1 806 tested weightings are therefore superseded by this chapter and are not repeated here. The board is recomputed under the partition, with Bory at 120, at the next canvas version (the owner, 9 November, R2-Q13). Until then the board carries a note that it predates version 4.

## 12.5 The futures map, in one paragraph

Chapter 9 describes it. Rows are the model's draws, each centred on the mean of the options, so the map reads relative performance; heights are the uncentred mean savings; the absolute question, which options fail together, is answered by the shared-loss matrix with its denominators (T8), not by the map. Layout convergence, representation fit and robustness to the declared assumption settings are three different things and are reported apart. A grouping the report keeps after a failed global

stability test (the German valley, the Czech–Polish ridge) is kept only if it is tested on its own; otherwise the relation matrix is printed without a spatial reading. "No site is a backup of another" is a claim about tested downside dependence between the focal sites and the rest; it says nothing about operational switching, which needs its own time, cost and feasibility assessment.

CHAPTER 13

# Grades, sources and the release gate

Every number in a type-2 report carries a grade in small type and a numbered source at the end of its chapter. The grade says how the number came to be; the source says where. Behind the one printed grade stand four attributes.

## 13.1 The exemplar's seven grades

| Grade | Meaning (R2-P04 § 3.5) | Exemplar instance |
|---|---|---|
| *z dat klienta* | a number read from a client's file or document | contract end 30. 6. 2030; 36 comparable plants |
| *výpočet* | a number the program computed from stated inputs in one computation: the base case, break-even, finance, operations | base-case saving 170,7 mil. EUR; T4–T6 |
| *odhad* | a range derived from client data, for instance from comparables; and every number from the futures, which carries the full label *odhad (metoda čeká na schválení metodikem)* while the method is unregistered | overrun median 17 %; every median, interval, probability, value of information, boundary, delay table and the futures map |
| *úsudek* | the author's assumption where the client has no answer; also proposed dates | the 20 % compensation chance; the correlations; the plan's dates |
| *veřejný zdroj* | a figure from public research, provisional until verified at the source | allowance price 89,10 EUR/t; exchange rates |
| *právní předpis* | a provision, likewise provisional until verified | tax rates, employer contributions, the directive's limits |
| *průzkum* | answers from a survey; type 2 does not use it | — |

## 13.2 The printed grade and the three attributes behind it

The printed grade says how the number came to be: its origin (*z dat klienta*, *veřejný zdroj*, *právní předpis*, *úsudek*, *průzkum*) or, for a model result, its method (*výpočet*, *odhad*). Three more attributes travel with every number and are never folded into the grade: its **lineage** (which inputs it was computed from and their grades), its **uncertainty** (the range, interval or probability basis, printed beside it) and its **validation** (the method in the register or awaiting the methodologist; a public input verified or *prozatímní*). Version 3 said a claim prints the weakest grade of the fields behind it and, in the same paragraph, that a number computed from a judgement prints *výpočet*; the two cannot both hold. Version 4 keeps the method grade on the number and carries the lineage separately: a base-case figure computed from the 20 % compensation judgement prints *výpočet*, and its lineage shows *úsudek*, which the sentence must state where the judgement is decisive.

| Printed grade | Who may print it | Admission rule (code) | Carried beside it |
|---|---|---|---|
| *z dat klienta* | a number read from a client file or document under its contract | CLIENT_REPORTED: the source locator resolves to the accepted contract and copy | validation *ověřeno*; lineage the file and row |
| *výpočet* | one deterministic computation from stated inputs: base case, break-even, finance, operations | CALCULATED: the run's artefact and the inputs' revision pinned | lineage: the inputs and the weakest grade among them; uncertainty: none of its own |
| *odhad* | a statistic of the futures, or a range derived from client data | ESTIMATED: the method's register status and the comparison's resolution state | validation: *(metoda čeká na schválení metodikem)* while unregistered; uncertainty: the interval, and *nerozhodnuto* where the bound crosses a threshold |
| *úsudek* | a person's assumption where the client has no answer | a labelled judgement with its author's name | lineage: none; uncertainty: the range the author stands behind |
| *veřejný zdroj* | a figure from public research | a typed input with quote, source and date | validation: *ověřeno* or *prozatímní* |
| *právní předpis* | a provision | a typed legal input (`legal-context#LGL5`) | validation as above; the effective date |
| *průzkum* | survey answers; not used in type 2 | the survey origins | — |

In the code the print grades are five today, `PrintGrade` in `domain/report/evidence.py`: measured, calibrated, modelled, holdout-pending, unknown, a missing grade printing "?". The plan adds CALCULATED and ESTIMATED with the register's status (`cost-model-study#CP8`, `#CP10`), ADR 0027 the client-reported origin, and `#CP8b` (**new**) the table above as code: one row per grade, Czech

label, admission rule, the attributes carried. Board MV4 prints the seven grades with their attributes and where each occurs in report 2; the result boards' chips show the printed grade.

### 13.3 Sources, numbered per chapter

- Each chapter ends with *Zdroje a stupně*: a numbered list; each printed number cites a list entry by its number (R2-P02: "z dat klienta 1").
- An entry names the document by its cited name and page or paragraph ("Zápis zadání s klientem §6", "Dopis dodavatele, s. 1"), the file for data, the table for a computed value ("výpočet: tabulka T5 v příloze"), or the method's status for an estimate ("odhad (metoda čeká na schválení metodikem)").
- An entry may carry two grades when a number rests on both ("výpočet; z dat klienta: Srovnatelné závody"); the weaker is printed at the number.
- The document register (Appendix A) maps every cited name to its file, so a source is checkable with one open.
- In AIA, the section-source manifest (`domain/report/sources.py`) already records each section's sources by artefact id and SHA-256; the numbered list is its printed form.

### 13.4 The provisional state

*Prozatímní* is a validation status, not a grade. It marks a public input not yet verified at the source, and it is printed in three places: the input table's state column (T9), the sentence that uses the input ("dlouhodobý vývoj cen energie (prozatímní údaj)"), and the first-page state box with the count. The count comes from the manifest (chapter 2): 16 provisional of T9's 67 rows. It is removed only by a dated verification by a named person (board MV2 § 02b). The report's running head carries it while any decisive input is provisional: the exemplar prints *Pracovní verze · prozatímní* on every page. The exemplar's text says 16 of 54 while its T9 has 67 rows; boards MV4, M6 and MV8 record that as the report's one K6 finding, and the manifest rule exists so that it cannot recur.

### 13.5 The release gate (R9, adopted 9 November 2026)

The owner adopted R9 on 9 November (R2-Q05): labelled unregistered estimates may appear internally and in an independently reviewed *předběžné* client draft; an approved recommendation needs registered methods, the independent review and every decisive comparison resolved. Document approval and recommendation eligibility are two states (chapter 2). This amends W27 (an unregistered estimate reaches the client report labelled, with a warning) and report-delivery RD0 of 22 October ("warnings for all of it") for the business-decision report: the warnings stay, and the recommendation's eligibility is a computed state beside them. The amendment is recorded in ADR 0046 by the docs PR that follows the scope's version 2.

- Every decision-critical claim is checked twice: support in its source (code) and correct interpretation (a person who is not the claim's author).
- A recommendation that rests on a method outside the register, or on a comparison still unresolved under the convergence rule, is *předběžné doporučení*. The exemplar's recommendation rests on the futures, so it is provisional in this sense, and its summary says so.
- Exploratory material enters the report only with explicit limits ("exploratory; does not support the recommendation"). The weighting-scenario map is such material.
- Approval records the exact version and the open qualifications; "approved document" means reviewed and releasable, "recommendation eligible" means the recommendation may be stated as approved. The second is never typed; code derives it from the register, the review record and the resolution states.
- The Deep Research thresholds of M-7 (citation precision ≥ 90 %, unsupported claims < 10 %) are one check, not the release criterion.

What exists today: approval once per stored version with the approver and self-approval rules recorded (`application/report_approval.py`); the client gate's checks as warnings recorded on the approval's ledger row (`domain/report/client_gate.py`, RD0); the sentence check (RD7). What R9 adds: the recommendation's eligibility as a computed field of the approval, the qualifications list, and the second, human reading of each decision-critical claim.

CHAPTER 14

# Governance

Reproducibility proves the code computes what is written. It does not prove the model is the right one for the decision. That needs a review by someone who did not accept the inputs, a register that says which methods are approved, and a record of what is still open and whose word closes it.

## 14.1 The validity review (R1)

Before a recommendation is approved, an independent reviewer answers five questions; the researcher who accepted the inputs answers to them, never reviews them (board MV8 § 01; review R1).

| # | Question | In the ChemVitro exemplar |
|---|---|---|
| 1 | Is every option represented comparably? | the same plant, horizon, rate and timeline; site-specific months and costs from the site's documents (R2-P02, Appendix B) |
| 2 | Are the material costs, benefits, constraints and implementation dependencies in the model? | the EPC exclusions priced; outside the model: transfer prices, hiring time in the schedule, raw-material freight, the annual step (R2-P15) |
| 3 | Could an excluded item reverse the recommendation? | raw-material freight would lower savings by 9–14 mil. EUR without changing the verdict (P1); the 2030 air limits could (R2-P01) |
| 4 | Do extreme inputs give sensible results? | seven of the report's own computations read as extreme-input tests on board MV8 § 03: compensation chance 0 % and 30 %, the 2025 price, the post-2035 premium at 0 % and 30 %, the investment read "in every future", a six-month delay without compensation; all seven behave as expected |
| 5 | Has an independent reviewer challenged the structure and the decisive assumptions? | not recorded in the exemplar; board MV8 (session E) records a reviewer who did not work on the study, assigned 5. 11., with three questions open: the compensation chance, the post-2035 terms, the investment reading |

With the questions, board MV8 holds a table of the ten limits the exemplar admits in its chapter 14, each with the check that catches it and its state: three caught (the map's failed stability test, the 91 kt outlier, the simplifications listed with their effect), seven open (the compensation judgement awaiting the authorities' document, the stations' distance from the sites, the sixteen provisional inputs, the 2030 directive, the version without correlations that did not converge, combinations not computed, the investment reading awaiting the CFO). Then the statement that validity is not reproducibility and the six criteria K1–K6 with their state on report 2: K1 open (two judgements without a document), K2 partial (combinations and convergence), K3 met, K4 waiting for the recomputation after verification, K5 met, K6 one finding. The review's record is part of the approval; its open items are the four qualifications that keep the recommendation *předběžné*.

## 14.2 Register entries the exemplar waits on

The Sociomap register (`domain/sociomap_register.py`) and Metodika's versions (`domain/methodology/`, ADR 0023) hold what is approved. A type-2 study needs entries that do not exist. Until each exists, its results print *odhad (metoda čeká na schválení metodikem)*. Board MV5 (session E) carries the same list as the register's rows for report 2, with what the methodologist signs for each, and keeps NPV, IRR, payback, break-even and the tornado as *v registru*: a single computation needs no entry.

| Method | What the entry must fix | Where it is used |
|---|---|---|
| Monte Carlo over the input table: PERT, declared correlations (Iman–Conover), empirical joint draws from comparables, yes/no events, mixtures | the default distribution by source; the correlation method; the stopping rule and its margin; the seed convention | every median, interval and probability (T1–T3, T5) |
| The decision boundary | the grid over an input's range; recomputation of the futures per value; how the flip is reported | R2-P06 § 5.3 |
| Value of information for the rule, and EVPPI | the partition of futures by the answer; the rule applied per partition; EVPPI by Strong and Oakley (2013); the order of the two measures; the enquiry table's columns | R2-P09 |
| The futures map (observers = futures) under `aia-sociomap-5` | the mode contract: which pair statistic, if any; the determinedness test; the stability test's settings, agreement measure and threshold; the relative and absolute readings (`sociomapa-systemu#SYS9`, `SYS9b`, `SYS12`) | R2-P12, T7, T8 |

| Method | What the entry must fix | Where it is used |
|---|---|---|
| The weighting-scenario map | the partition's admission test (splitting invariance), value anchors and swing weights, the fans' concentrations and k, score bounds and the robust-leader rule, shares of tested settings per scenario, the view's mode contract (`sociomapa-systemu`#SYS13, SYS12) | board MV9 |
| Estimates from comparables | the filter, the statistic, the minimum sample, how a joint draw is recorded (`cost-model-study`#CR3b; `client-data-in-analysis`#12, an estimate from client rows as a Metodika method) | overrun and delay, pre-treatment, spur costs (T9) |

## 14.3 The methodologist's open questions

ADR 0046 lists ten (§ Open: multiple testing on maps; levels or changes; effective sample size; resampling for 24–36 rows; the surrogate null; the relation statistic; the synthetic suite's thresholds; the discount rate in probabilistic rules; PERT inputs; forecast evaluation), with board MV5's recommendation for each. The exemplar and the scope (§ 8) add eight; board MV5 § 02 asks six of them with a recommendation each (PERT inputs; input correlations by Iman–Conover, a Gaussian copula or the common-plant draw; the stopping rule; the discount rate; whether pair statistics are read on the futures map; the stability threshold) and defers the unit and period modes' questions:

1. **The futures map's contract** (R5, `sociomapa-systemu`#SYS9): rows are 20 000 chosen draws, so what does it print instead of the respondent method's "2 of 28 reliable" and split-half 0,91 against 0,59 (R2-P12 § 11.2)? The exemplar drew links by the `-5` pair test with the false-discovery correction; board MV3's scenario contract says no null applies when the model creates the relations.
2. **The stability test** (R2-P18 B): Spearman on the 28 distances between every two of three correlation settings, threshold 0,8; approve, or replace the statistic and the threshold. The "strong" setting (0,95 / 0,85 / 0,4) is the author's judgement.
3. **The empirical joint draw** (`cost-model-study`#CR3b): one comparable plant per future carrying overrun and delay together, as an alternative to PERT with a declared correlation; whether 36 plants support it and how the 91 kt outlier is treated.
4. **The convergence rule and the endpoint semantics of chapter 2**, adopted as the baseline on 9 November: confirm or amend the doubling from 20 000 to 1,28 million, the repeated-look exact binomial bounds, the 5 % error budget, quantiles fitted as quantiles and hard bounds as support; PERT λ; and when a mixture or a uniform is the better default for the 22 inputs the files do not hold.
5. **Dependence**: the contract recommends a Gaussian copula for continuous inputs (validated rank-to-latent conversion, a valid latent correlation matrix, preserved empirical pairs, explicit event dependencies) in place of sample-wide Iman–Conover reordering, so the numerical error contract matches the sampler; confirm, and set the tail stresses and the achieved-dependence check.
6. **Value of information for the rule** (`cost-model-study`#CR6b, R7): the rule re-applied per declared answer branch with the enquiry's cost, duration and shared dependencies inside the comparison; approve the definition, including its meaning when the rule's choice is "nothing", how it combines across several enquiries (the exemplar's 152,1 for all three authorities), and the estimator checks on the fixture O-INFO (17,5 for the rule, 12,5 as expected-benefit EVPI; § 14.6).
7. **The weighting map under the partition** (chapter 12): the value functions' anchors per perspective, the swing-weight elicitation, the fans at concentrations 20, 60 and 180 with 300 settings each, the robust-leader rule on score intervals, and the map-fit thresholds of its mode contract; shares are shares of tested settings, per scenario, never pooled (R4).
8. **The seven print grades mapped to the code's** (`cost-model-study`#CP8b, Q1): one table, Czech labels, admission rules.
9. **A register entry** for `aia-sociomap-5` with futures or settings as observers (today's entry covers respondents' ratings), or a separate method id per mode (`sociomapa-systemu`#SYS12).

## 14.4 The owner's decisions of 9 November, and what is still open

| Id | Decision | Where it stands |
|---|---|---|
| R2-Q01 | The ChemVitro fixture reads the investment condition as total base-case installed investment in 2026 EUR; Aram confirms it as the case's author, standing in for the CFO. | decided (a); the rule is frozen for the fixture once his word is recorded |
| R2-Q02 | The comparables' cost basis: a versioned clarification of the file's footnote on a like-for-like basis, or two bounded interpretations with conditional conclusions. | **open**; recommended: the clarification (chapter 6) |
| R2-Q03 | The perspective weights weigh a disjoint cost partition, one weight vector for all sites, NPV printed beside. | decided (a); chapter 12 |
| R2-Q04 | Strategic and marketing have no leaf for ChemVitro: visible at weight 0, or research evidenced objectives now. | **open**; recommended: weight 0 for this case |
| R2-Q05 | R9: qualified *předběžné* drafts after independent review; approved recommendations only on registered methods and resolved comparisons; two states. | decided (a); chapters 2 and 13; the ADR 0046 amendment follows in the docs PR |

| Id | Decision | Where it stands |
|---|---|---|
| R2-Q06 | The researcher records client-approved weight vectors through the existing approval flow. | decided (a) |
| R2-Q07 | Three build sessions, one analyst session that never writes model code, the methodologist one day a week. | decided (a); the weekly day is asked of the methodologist |
| R2-Q10 | The seven contract defaults as the baseline for the methodologist's review. | decided (a); chapter 2 |
| R2-Q11 | Acceptance: the independent numerical oracle first, the ChemVitro answer key second. | decided (a); § 14.6 |
| R2-Q12 | A separate session writes the oracle; the methodologist signs the fixtures. | decided (a) |
| Q1 | The attributes of a number as the vocabulary of the outline and the report | the legend on M6 and the labels on MV4; the mapping table is `CP8b` (chapter 13) |
| Q2 | Enforce data classes before a client's data flows into a model | yes for Class A in staging and production, recommended |
| Q4 | The SOMECS name and licence | until checked, the map is *Sociomapa systémů* |

Settled by the brief of 8 November: the case for the review's step 5 is ChemVitro, and the MVP's spine is type 2 end to end on it. Settled by the scope: the type-2 model is a model type of its own, `site_cost_of_supply` (`cost-model-study#CM8`), inside the one merged study (`#CP3`, A2: no new study kind). The Velis boards M4c and M4e stay on the canvas as parked.

## 14.5 Change propagation (K4)

A revised input is a new revision of the input table. Every result, table, map and approval that read the old revision is marked stale until someone reviews it; the report's version carries the revision it was composed from. In AIA today the pipeline's fingerprints and the impact rule (`domain/pipeline.py`, the research impact route) do this for a research design; the model platform reuses the rule (`cost-model-study` CP5: inputs frozen by artefact id and SHA-256).

## 14.6 Acceptance: the oracle first, ChemVitro second

A pipeline could pass the answer key with wrong cash flows: find the planted facts, add a charge for each trap, apply the rule faithfully to incorrect numbers, print the caveats, reproduce the same bytes. So acceptance has two layers (the owner, 9 November, R2-Q11). An analyst session that never writes the model code, and never imports it, freezes equations, inputs, expected intermediates, tolerances and conventions; the methodologist signs them (R2-Q12).

| Fixture | Content | Expected |
|---|---|---|
| **O-CF** | two years, end-year flows, 10 % discount, no inflation, tax, residual or uncertainty, toy units. Buying: 0 capital, 100 a year. A: 50 capital, 60 a year. B: 20 capital, 80 a year. $C = I + c_1/1{,}1 + c_2/1{,}1^2$; saving = $C_{buy} - C_{site}$. | present cost 173,554 / 154,132 / 158,843; saving 0 / 19,421 / 14,711; at zero discount 30 and 20; a terminal receipt of 12,1 at year 2 lowers A's present cost by exactly 10,000. A separate dated-flow fixture tests the ChemVitro convention (mid-year, inflation); this toy does not prescribe it. |
| **O-INFO** | two equally likely states; A saves 40 / −20, B saves 5 / 5, buying 0; start and investment gates pass; P(beats buying) ≥ 0,75 and P(loss > 20) ≤ 0,10 as gates. | without information the rule picks B (A fails the gate at 0,50): value 5. Perfect information: A in the good state, B in the bad: 22,5. Value for the rule 17,5. Expected-benefit optimisation picks A at 10, so its perfect-information value is 12,5. Both are reproduced and told apart. |
| **O-CHEM** | the ChemVitro configuration frozen after the cost-basis reconciliation: file hashes, transformations, the manifest, equations, event precedence, the shared and site shock structure, the rule and the method versions; investment, timing and site cash-flow checkpoints computed independently before production results are compared. | the reference reading (Kamenec 39,8 with compensation only where a station exceeds the limits) and the exemplar reading (137,1) frozen as two configurations; neither number is a target for the other. |

The tests the fixtures feed: units, currency and periods (the kilogram row hand-normalised to 2 844 t, the USD row at a frozen dated rate, a deliberately misclassified non-hazardous quote; silent substitution fails); accounting identities and scope (every expense once; zero usable tax capacity gives no immediate tax benefit); criterion splitting (a logistics line split with its total conserved leaves scores, ranks and shares unchanged; scoring parent and children together fails admission); missing-data states (known, evidenced zero, globally irrelevant, unknown range, several unknowns that jointly reverse a ranking; no site-specific reweighting); threshold uncertainty (known probabilities below, at and above a threshold; samples around 0,797 and 0,804; the whole stopping procedure under repeated seeds, not one final interval); correlated information branches (a finite joint-state table with impossible answer combinations; an independent-marginal substitution fails); distributions and dependence (analytic PERT moments, quantile-fitted marginals, paired observations, common national and site shocks); maps (known matrices, identical profiles, a dominated option with a similar profile, poor geometry, deliberate instability: similarity never becomes a co-winning claim); downstream invalidation (every stored root mutated, descendants stale, old artefacts immutable); claim meaning and release (wrong percentile wording,

rounding that hides a gate failure, an unregistered method, an unresolved comparison stated as a verdict). The integration exit is one real browser, API and worker journey on the case files that produces the T1–T9 equivalents, both maps, the enquiry table and the QA annex from frozen artefacts and passes the independent review.

### 14.7 The independent review's evidence

Recommendation eligibility (chapter 2) needs a review record, not a reviewer's name. The record holds: the five questions of § 14.1 answered with the evidence each rests on; the limits the report admits with the check that caught each or the reason it is still open; the extreme-input tests actually run (the exemplar's chapter 14 lists seven computations that read as such tests, board MV8 shows them, and FND14 builds four; the register entry says how many a type-2 study must run and which); the oracle fixtures' pass record with the signer's name; the resolution state of every decisive comparison; and K1–K6 each marked met, partial or open with its anchor. A review with an open item leaves the document approvable as *předběžné* and the recommendation ineligible.

APPENDIX A

# Glossary

| Term | Meaning in this methodology |
|---|---|
| Base case (*základní případ*) | one computation with every input at its likely value; compares options and feeds sensitivity; not an estimate of the saving |
| Baseline | the option every other is compared with; for ChemVitro, keep buying |
| Leaf | one cash-flow line (or lines that always travel together), discounted, allocated to exactly one perspective; a perspective's total is the sum of its leaves |
| Data contract (*datová smlouva*) | which columns of a client file are used and how; units, currency, price year, personal columns, period; accepted by a person |
| Decision boundary (*hranice rozhodnutí*) | what the rule chooses as one assumption moves over its range, the futures recomputed for each value |
| Disposition | what was done with a data finding: used; corrected with the reason; left out; taken as an assumption with a range |
| EVPPI | expected value of partial perfect information: the gain in mean outcome from knowing one input before deciding (Strong and Oakley 2013) |
| Fan | k perturbed weight vectors around a named weighting scenario, from a declared concentration; a sensitivity setting, not a probability |
| Futures (*budoucnosti*) | the seeded draws of all inputs at once; probabilities are shares of them |
| Futures map (*mapa budoucností*) | a Sociomap whose observers are the futures and whose objects are the options; centred per future |
| Gate | a non-compensable requirement; fail or unknown blocks the recommendation |
| Grade (*stupeň*) | the small-type label on a printed number: how it came to be; chapter 13 |
| Guarantor (*stojí za ním*) | the person who stands behind an input's range |
| Interpretation record | a reading of a rule's condition that the rule does not fix, with the alternative, its effect and who confirms |
| Mode contract | per Sociomap mode: what a row is, what the claim is about, what is measured, what the null means, the resampling, what the stability measures |
| Oracle fixture | a case with independently known answers (O-CF, O-INFO, O-CHEM) frozen by an analyst who does not write the model code and signed by the methodologist; the pipeline must reproduce it before the ChemVitro answer key counts |
| Outline (*Osnova zprávy*) | the report's claims in the order of the argument, each with its source and why it matters, before any text |
| Perspective | a set of leaves: financial, strategic, operational, marketing/commercial, logistics, environment and permitting, people; one weight per perspective, the same for every option |
| Recommendation eligibility | the computed state that lets a recommendation be stated as approved: registered methods, the independent review closed, every decisive comparison resolved; apart from document approval |
| Repeated-look bound | a confidence bound still valid after the draws have been doubled and checked several times; the convergence rule uses exact binomial bounds of this kind |
| Provisional (*prozatímní*) | a public input not yet verified at the source |
| Register (*registr metod*) | the methods the methodologist has approved, each at its version and fingerprint |
| Rule (*pravidlo*) | the client's decision rule, written before any result: the measure, the choice, the conditions with thresholds |
| Shared failure | when one option is worse than the baseline, the share of those futures in which another is worse too (T8) |
| Share of tested settings | the share of a fan's weight vectors under which an option is uniquely first, with the shares of ties and of unresolved settings beside it; per scenario, never pooled; not a probability |
| Stability test | the map recomputed under settings declared in advance; stable only if the distance orders agree above the threshold |
| Stopping rule (convergence) | how many futures to run: 20 000, doubled at declared checkpoints up to 1,28 million, until every gate and decisive comparison clears its repeated-look bound; a comparison that does not prints as unresolved |
| Unresolved (*nerozhodnuto*) | a comparison whose bound still crosses its threshold at the draws run so far; printed as an interval, never as pass or fail |
| Value for the rule (*hodnota pro pravidlo*) | what knowing an answer in advance and applying the rule again would bring, averaged over the possible answers |
| Weighting-scenario map (*mapa scénářů vah*) | a Sociomap per scenario whose rows are tested weight settings and whose objects are the options; height is the score under the selected setting; preference information, with the gates beside it and the rule's verdict in its own panel |

APPENDIX B

# The exemplar in numbers

Condensed from the exemplar's tables (R2-P18). Every value is the exemplar's; the grades are its grades. NPV 2027–2044, mil. EUR, pre-tax, 8 % unless stated. odhad marks a number from the 20 000 futures (method awaiting approval); výpočet a base-case computation.

## B.1 T1 · The rule at 8 %

| Site | Median saving | 90 % interval | P(beats) ≥ 0,75 | P(loss > 20) ≤ 0,10 | P(start by 1. 7. 2030) ≥ 0,80 | Investment ≤ 175 | Passes |
|---|---|---|---|---|---|---|---|
| Vrbice (CZ1) | 110,8 | −40,7 … 331,5 | 0,88 ✔ | 0,08 ✔ | 0,21 ✘ | 160,6 ✔ | no |
| Lipová (CZ2) | 126,0 | −30,2 … 349,4 | 0,91 ✔ | 0,06 ✔ | 0,76 ✘ | 158,6 ✔ | no |
| Kamenec (CZ3) | 132,4 | −23,5 … 356,9 | 0,92 ✔ | 0,05 ✔ | 0,76 ✘ | 152,3 ✔ | no |
| Wendelstedt (DE1) | 82,3 | −74,7 … 308,1 | 0,80 ✔ | 0,15 ✘ | 0,94 ✔ | 171,8 ✔ | no |
| Lindenau (DE2) | 86,5 | −68,8 … 311,7 | 0,81 ✔ | 0,14 ✘ | 0,77 ✘ | 170,8 ✔ | no |
| Eschfeld (DE3) | 61,2 | −94,4 … 285,7 | 0,73 ✘ | 0,21 ✘ | 0,67 ✘ | 175,5 ✘ | no |
| Odrzyca (PL1) | 134,7 | −23,8 … 359,1 | 0,92 ✔ | 0,05 ✔ | 0,75 ✘ | 148,9 ✔ | no |
| Bory (PL2) | 123,2 | −30,9 … 344,4 | 0,91 ✔ | 0,06 ✔ | 0,19 ✘ | 143,9 ✔ | no |

Median, interval and probabilities odhad; investment in the base case výpočet. At 6 % and 10 % the verdict is the same (T2, T3): the deadline probability does not depend on the rate. Keep buying costs 897 mil. EUR in the median of the futures and 911,6 in the base case.

## B.2 T4 · Finance, base case

| Site | Pre-tax 6 % | 8 % | 10 % | Post-tax 8 % | Median post-tax 8 % | IRR | Payback | Start (base) |
|---|---|---|---|---|---|---|---|---|
| Vrbice | 189,2 | 134,5 | 92,7 | 96,6 | 87,5 | 18,2 % | 2035 | 2030,92 |
| Lipová | 219,3 | 159,9 | 113,8 | 116,3 | 99,9 | 19,2 % | 2034 | 2029,92 |
| Kamenec | 225,6 | 165,9 | 119,6 | 121,3 | 107,3 | 19,9 % | 2034 | 2029,92 |
| Wendelstedt | 149,8 | 99,8 | 61,2 | 84,3 | 72,1 | 14,8 % | 2035 | 2029,92 |
| Lindenau | 156,0 | 105,7 | 67,0 | 92,4 | 80,0 | 15,5 % | 2035 | 2030,17 |
| Eschfeld | 139,4 | 91,2 | 54,2 | 69,3 | 45,9 | 14,4 % | 2036 | 2030,25 |
| Odrzyca | 230,6 | 170,7 | 124,2 | 118,6 | 107,2 | 20,5 % | 2034 | 2029,92 |
| Bory | 200,5 | 145,8 | 103,9 | 99,7 | 96,0 | 20,1 % | 2035 | 2031,17 |

Base case výpočet; the post-tax median odhad. Tax rates provisional. Incentives are not in the base case; the futures grant them with probability 0,5.

## B.3 T6 · Operating cost per tonne in 2032, base case, EUR/t

| Site | $CO_2$ | Freight | Energy | People | Waste | Raw materials | Water | Maintenance, park | Total |
|---|---|---|---|---|---|---|---|---|---|
| Vrbice | 56 | 38 | 179 | 121 | 17 | 760 | 26 | 113 | 1 311 |
| Lipová | 56 | 20 | 179 | 116 | 20 | 760 | 25 | 124 | 1 300 |
| Kamenec | 56 | 24 | 179 | 124 | 16 | 760 | 26 | 111 | 1 295 |
| Wendelstedt | 56 | 58 | 182 | 203 | 13 | 760 | 25 | 157 | 1 455 |
| Lindenau | 56 | 53 | 190 | 195 | 15 | 760 | 32 | 132 | 1 434 |
| Eschfeld | 56 | 55 | 190 | 232 | 14 | 760 | 34 | 122 | 1 463 |
| Odrzyca | 56 | 37 | 192 | 102 | 18 | 760 | 21 | 105 | 1 292 |
| Bory | 56 | 33 | 192 | 109 | 17 | 760 | 20 | 103 | 1 290 |

výpočet. A bought tonne would cost about 2 340 EUR in 2032 in the base case, premium and freight included.

## B.4 T7 · Relations between sites across the futures

| | Vrb | Lip | Kam | Wen | Lin | Esch | Odr | Bory |
|---|---|---|---|---|---|---|---|---|
| Vrbice | – | 0,50 | 0,52 | −0,27 | −0,27 | −0,26 | −0,15 | −0,07 |
| Lipová | 0,50 | – | 0,55 | −0,22 | −0,23 | −0,27 | −0,14 | −0,15 |
| Kamenec | 0,52 | 0,55 | – | −0,23 | −0,23 | −0,26 | −0,14 | −0,15 |

| | Vrb | Lip | Kam | Wen | Lin | Esch | Odr | Bory |
|---|---|---|---|---|---|---|---|---|
| Wendelstedt | −0,27 | −0,22 | −0,23 | – | 0,56 | 0,70 | −0,12 | −0,17 |
| Lindenau | −0,27 | −0,23 | −0,23 | 0,56 | – | 0,71 | −0,12 | −0,15 |
| Eschfeld | −0,26 | −0,27 | −0,26 | 0,70 | 0,71 | – | −0,15 | −0,15 |
| Odrzyca | −0,15 | −0,14 | −0,14 | −0,12 | −0,12 | −0,15 | – | 0,48 |
| Bory | −0,07 | −0,15 | −0,15 | −0,17 | −0,15 | −0,15 | 0,48 | – |

`odhad`; `aia-sociomap-5`, the judgement's correlations, centred per future. Two of 28 pairs reliable. Split-half 0,91 against 0,59. Stability: 0,95 / 0,84 / 0,75 between the three settings, the last under 0,8: failed.

## B.5 T8 · Shared failure of the two leading sites

| When this site is worse than buying … | Vrb | Lip | Kam | Wen | Lin | Esch | Odr | Bory |
|---|---|---|---|---|---|---|---|---|
| Odrzyca | 90 % | 84 % | 79 % | 98 % | 97 % | 99 % | – | 91 % |
| Kamenec | 99 % | 93 % | – | 98 % | 98 % | 100 % | 75 % | 83 % |

`odhad`; shares of the futures, uncentred. No site is a backup of another.

## B.6 The decision boundary and the input table in brief

| Compensation chance | Rule chooses | P(on time) Lipová / Kamenec / Odrzyca |
|---|---|---|
| 0 % | Odrzyca | 0,942 / 0,942 / 0,942 |
| 10 % | Odrzyca | 0,852 / 0,851 / 0,844 |
| 14 % | Odrzyca | 0,817 / 0,813 / 0,808 |
| 15 % | Kamenec | 0,807 / 0,804 / 0,797 |
| 16 % | nothing | 0,797 / 0,795 / 0,787 |
| 20 % (the model) | nothing | 0,760 / 0,759 / 0,748 |

`odhad` (R2-P06 § 5.3). At 15 % the two probabilities nearest the threshold, 0,804 and 0,797, are unresolved on 20 000 draws under the convergence rule (chapter 2); the column reads "Kamenec, unresolved" until the draws clear it. T9 holds 67 rows (the exemplar's text says 54): 29 read from client data (*ověřeno*), 16 public inputs awaiting verification (*prozatímní*), 22 judgements (*úsudek*). Overrun and delay: median 17 %, 90th percentile 30,2 %, delay median 3 months, from 36 comparable plants, drawn together.

APPENDIX C

# Board index

The boards of the Design canvas *AIA Obchodní model* at version 66 (8 November), in the order of its rows, with the session that changed each. Where version 4 of this document moved past a board, the row says what the next canvas version changes. "Landed" means the session's branch is pushed and this document is aligned to it; "in progress" means the session is still working and the chapters that cite the board will be aligned when it lands. Boards marked parked stay on the canvas as planned, not deleted. Since 8 November a board that carries ChemVitro does so beside its Velis sections, each section marked with its case.

| Code | Board | File | Session | State after 8 November |
|---|---|---|---|---|
| Z0 | *Zadání: průchod* | `Z0Flow.dc.html` | A | landed: the flow and the board list around the spine |
| Z1 | *Zadání: začátek* | `Z1Start.dc.html` | — | unchanged |
| Z2 | *Zadání: triáž* | `Z2Triage.dc.html` | — | unchanged |
| Z3 | *Zadání: rozhodnutí a pravidlo* | `Z3Decision.dc.html` | B | landed: § 03b the type-2 rule with four conditions as gates and the interpretation record |
| Z4 | *Zadání: analýzy do zprávy* | `Z4Analyses.dc.html` | — | unchanged |
| Z5 | *Zadání: obchodní model* | `Z5Model.dc.html` | B | landed: § 04b the parts of the cost-of-supply model, the timeline, the gate, the views, the perspectives |
| Z7 | *Zadání: připravenost a předání* | `Z7Ready.dc.html` | — | unchanged |
| M1 | *Model: Data* | `KM1Data.dc.html` | B | landed: § 01b–03b the document register, what the model takes with file and row, the 42 findings with dispositions |
| MV1 | *Data: datová smlouva* | `MV1Contract.dc.html` | B | landed: § 01b–03b the contract and quality report of the purchases file, units and currencies as corrections |
| MV2 | *Rešerše: typované vstupy* | `MV2PublicData.dc.html` | B | landed: § 01b the sixteen provisional inputs, § 02b verification at the source as a step |
| MV2b | *Rešerše: zjištění podle analýz* | `MV2bFindings.dc.html` | — | unchanged |
| M2 | *Model: Předpoklady* | `KM2Assumptions.dc.html` | B | landed: § 01b T9 (67 rows, the K6 finding), § 02b uncertainty types, § 03b the estimate from comparables, § 03c the scope check, § 03d correlations and the stability test declared |
| M3 | *Model: Model* | `KM3Model.dc.html` | C | landed: the shared timeline and the eleven items of the cost of supply |
| M4 | *Model: Výsledky* | `KM4Results.dc.html` | C | landed: T1 at 8 %, T2 and T3, the decision boundary, delays, tornado, break-even, value of information |
| M4b | *Výsledky: Business case a finance* | `KM7Finance.dc.html` | C | landed: T4 and T5 with P(investment ≤ 175) |
| M4c | *Výsledky: Účetnictví* | `KM8Accounting.dc.html` | C | parked for type 2 |
| M4d | *Výsledky: Operační výzkum* | `KM9Operations.dc.html` | C | landed: T6 with each column named as its component |
| M4g | *Výsledky: Harmonogram* | `KM12Schedule.dc.html` | C | new, landed: the start against the deadline, what lengthens each tail, delay × hiring as one grid |
| M4f | *Výsledky: Logistika* | `KM11Logistics.dc.html` | C | landed: product freight per site, the Odrzyca spur, the Eschfeld bridge, the trucking surcharge, the logistics rows of T9 |
| M4e | *Výsledky: Velikost trhu* | `KM10Market.dc.html` | C | parked for type 2 |
| MV3 | *Výsledky: Sociomapa systémů* | `MV3SystemMap.dc.html` | D | landed: the five readings of one engine, the futures contract, the exemplar's map in three settings, T7, T8, relative against absolute, the stability test and its verdict |

| Code | Board | File | Session | State after 8 November |
|---|---|---|---|---|
| MV9 | *Sociomapa: mapa scénářů vah* | `MV9ScenarioWeights.dc.html` | D | landed 8 November under version 3's two families (20 components with roles, six scenarios, 1 806 tested weightings on the real engine); superseded by chapter 12's partition and recomputed at the next canvas version, with Bory at 120 |
| SI1–SI4 | *Sociomapa z indikátorů*: method, matrix, options map, suppliers map | `SI1Method` … `SI4Suppliers.dc.html` | D | landed: cut to the method (evidence states and gates, swing weights, when the map may influence a recommendation); the overlap check they describe is replaced by the partition at the next canvas version |
| MV4 | *Osnova zprávy* | `MV4XRay.dc.html` | E | landed: the 18 parts R2-P01…P18 with typed records, the grades with their attributes, the argument's five parts mapped, the weighting map in part 3 |
| MV8 | *Zpráva: kontrola platnosti modelu* | `MV8Validity.dc.html` | E | landed: the ten limits of chapter 14 (3 caught, 7 open), the five questions, seven extreme-input computations, K1–K6 |
| M6 | *Model: Zpráva* | `KM6Report.dc.html` | E | landed: report 2 as rendered, *Jiné vážení* in the summary, the release state with four qualifications, today against new in the renderer |
| MV0 | *Definované MVP: přehled* | `MV0Scope.dc.html` | A | landed: rewritten around the spine; says "22 skórovaných složek" and words the rule as four conditions including the median; both corrected at the next canvas version (20; the objective and four conditions) |
| MV5 | *Metodika: validační sada a otázky* | `MV5Validation.dc.html` | E | landed: the register entries report 2 waits on, six questions, generators G7–G10, the release rule |
| MV6 | *Katalog analýz* | `MV6Catalogue.dc.html` | — | unchanged; the catalogue beyond the perspectives is parked |

Sizes and positions: `canvas.json` on each session's branch; the coordinator merges them. Reference renders of version 66: `reference/`. The canvas is published by the coordinator from the exported files; no session publishes it.